

Lectures on Computational Probabilistics

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Contents

1	Basic sampling methods	5
1.1	Generalized Inverse Method	6
1.2	Simulation of Gaussian distributions	6
1.3	Reject Sampling	7
2	MCMC: Monte-Carlo Markov Chains	9
2.1	Motivating Examples	9
2.1.1	Boltzmann-Gibbs measure	9
2.1.2	Ising model	9
2.1.3	Spin glass: a generalization	10
2.2	Backup Knowledge: Markov Chains	10
2.2.1	Basic concepts	10
2.2.2	Invariant and stationary measures	11
2.3	MCMC algorithms	12
2.3.1	Birkhoff's algorithm	12
2.3.2	Glauber dynamics of Ising model	12
2.3.3	Metropolis-Hasting algorithm	13
2.4	Distance on measures	14
2.4.1	Total variation	14
2.4.2	Wasserstein distance	14
2.4.3	More distances	14
3	Concentration and Functional inequalities	15
3.1	Confidence interval	15
3.1.1	CLT	15
3.1.2	Hoeffding's Theorem	15
3.2	Efron-Stein inequality and McDiarmid Theorem	17
3.3	Supplement: Spectral properties of Markov Chains	19
3.4	Correction of preceding proofs	20

4	Extreme Value Theory and Importance Sampling	23
4.1	Importance sampling	23
4.2	Extreme value theory	25
4.2.1	Gumbel: light tail	
	$G(x) = \exp(-e^{-x})$	25
4.2.2	Frechet: heavy tail	
	$G(x) = \begin{cases} 0 & x \leq 0 \\ \exp(-x^\alpha) & x > 0 \end{cases}, \alpha > 0$	26
4.2.3	Weibull: null tail	
	$G(x) = \begin{cases} \exp(-(-x)^\alpha) & x \leq 0 \\ 1 & x > 0 \end{cases}, \alpha > 0$	26
4.3	Extreme value via Poisson point process	27
5	Universality and condensation	29
5.1	Warm-up examples	29
5.2	A deeper version of CLT	30
5.2.1	Cauchy distribution	31
5.2.2	Asymmetric case	32
5.3	Proof of Theorem 5.0.1	32
5.4	Stable law	32
6	Power of Two Choices	34
6.1	Warm-up	34
6.2	Uniform allocation	35
6.3	Power of two choices	37
7	Random graphs: Erdos-Renji graph; preferential attachment	40
7.1	Preferential attachment	40
7.2	Erdos-Renji graph	41
8	Simulation of SDE	44
8.1	Brownian motion	44
8.2	Ito integral and Ito formula	45
8.3	SDE simulation	46
9	Langevin dynamic	47
9.1	Fokker-Plank equation	47
9.2	Poincare inequality	48
9.3	Log-Sobolev inequality	51
9.4	Inequalities about entropy	52

10 Diffusion models	54
10.1 Forward SDEs	54
10.2 Reverse of time	56
10.3 Score matching trick	58
11 JKO Schema	61
11.1 Flow (Gradient Flow)	61
11.1.1 Numerical Implementation	61
11.2 Heat Equation	62
11.3 JKO Schema (Jordan–Kinderlehrer–Otto) — Another variational formula	64

Notations

$X \stackrel{d}{\sim} Y$

i.i.d

$X_n \Longrightarrow Y$

$\mathbb{1}_A$

$\mathbb{E}[X]$

X and Y have the same distribution

independent identically distributed

X_n converges in distribution to Y

indicator for event A

expectation of random variable X

Lecture 1

Basic sampling methods

It should be enlightening to start with the famous experiment of Buffons', in which he tossed needles lengthed a uniformly randomly over several parallel lines on a plane with spacing l , and estimated the value of π based on the results.

Let X_N denote the number of tosses in which the needle touched a line. We claim that

$$\widehat{\pi}_N = \frac{2aN}{lX_N} \longrightarrow \pi \text{ as } N \rightarrow \infty$$

The proof is straightforward. As shown in the graph below, define h as the distance between the upper point and the line right below it, and define θ as the angle pointing from the direction of the needle to the direction perpendicular to the lines.

Then by the uniformity of the tossings, h is distributed uniformly over $[0, l]$ and θ is uniform over $[-, 2\pi]$. The case of touching the line can be determined by either $h - a \cos \theta < 0$ or $h + a \cos \theta > l$. By symmetry, obviously these two cases have the same probability. Hence

$$\mathbb{P}(\text{needle touching the line}) = 2\mathbb{P}(h - a \cos \theta < 0) \quad (1.1)$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \int_0^l \frac{1}{l} \mathbb{1}_{\{h - a \cos \theta < 0\}} dh d\theta \quad (1.2)$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \frac{1}{l} a \cos \theta d\theta = \frac{2a}{2\pi l} \quad (1.3)$$

By Law of Large Numbers,

$$\frac{X_N}{N} \longrightarrow \frac{2a}{\pi l} \text{ a.s. as } N \rightarrow \infty$$

Hence the claim holds.

What we have done above, in short words, is to build a Bernoulli distribution with parameter related to π , by using uniform distributions on h and θ . In some sense, this shows the fundamental significance of uniform distributions in this chapter.

Usually, we refer to the process of generating random variables with given distributions as *sampling*.

1.1 Generalized Inverse Method

Proposition 1.1.1.

If X is a random variable with increasing and right-continuous distribution function $F(x) = \mathbb{P}(X \leq x)$. Let $F^{-1}(u) = \inf \{x : F(x) \geq u\}$, and let U be a random variable with uniform distribution over $[0, 1]$, then

$$F^{-1}(U) \stackrel{d}{\sim} X$$

Proof. For any x ,

$$\mathbb{P}(F^{-1}(U) \leq x) = \mathbb{P}(F(x) \geq U) = F(x)$$

□

Corollary 1.1.2.

If F is continuous, then $F(x) \stackrel{d}{\sim} U$.

Proof. For any $g \in C_b(\mathbb{R})$,

$$\mathbb{E}[g(U)] = \int_0^1 g(u) du \tag{1.4}$$

$$= \int_{\mathbb{R}} g(F(x)) dF(x) \tag{1.5}$$

$$= \mathbb{E}[g(F(x))] \tag{1.6}$$

□

Example 1.1.3 (Exponential distributions).

For $X \sim \exp(\lambda)$, $F(x) = 1 - e^{-\lambda x} \mathbf{1}_{x>0}$, then $F^{-1}(u) = -\frac{1}{\lambda} \ln(1 - u)$ for $u \in [0, 1]$. Hence $X \sim \frac{1}{\lambda} \ln(U)$ where $U \sim \text{unif}[0, 1]$.

Remark 1.1.4.

This is how we sample an exponential distribution from uniform distributions, and also how we sample many other 1-dimensional random variables.

1.2 Simulation of Gaussian distributions

A natural thought would be to use the well-known Central Limit Theorem. Take i.i.d. X_i with $\mathbb{E}[X_i] = m$, $\text{Var}[X_i] = \sigma^2$, then

$$\frac{1}{\sigma\sqrt{n}} \sum_{i=1}^n (X_i - m) \implies \mathcal{N}(0, 1)$$

which implies that

$$\begin{aligned}\mathbb{P}\left(\frac{1}{\sigma\sqrt{n}}\sum_{i=1}^n(X_i - m) \leq x\right) &\xrightarrow{n \rightarrow \infty} \int_{-\infty}^x e^{-\frac{y^2}{2}} dy \\ \mathbb{P}\left(\frac{S_n}{n} \leq n + \frac{x\sigma}{\sqrt{n}}\right) &\xrightarrow{n \rightarrow \infty} \int_{-\infty}^x e^{-\frac{y^2}{2}} dy\end{aligned}$$

The Box-Muller method gives higher accuracy by generating normal distributions with 2 independent uniform distribution over $[0, 1]$.

Proposition 1.2.1 (Box-Muller).

If random variables R and θ satisfy $R^2 \sim \exp(\frac{1}{2})$, $\theta \sim \text{unif}[0, 2\pi]$, and $R \perp \theta$, then

$$(R \cos \theta, R \sin \theta) \stackrel{d}{\sim} \mathcal{N}(0, I_2)$$

Proof. Suppose random variables $(X, Y) \stackrel{d}{\sim} \mathcal{N}(0, I_2)$. For any $f \in C_b(\mathbb{R}^2)$,

$$\mathbb{E}[f(X, Y)] = \int_{\mathbb{R}^2} f(x, y) \frac{1}{2\pi} e^{-\frac{x^2+y^2}{2}} dx dy \quad (1.7)$$

$$= \int_a^{+\infty} \int_0^{2\pi} \frac{1}{2\pi} e^{-\frac{r^2}{2}} f(r \cos \theta, r \sin \theta) r dr d\theta \quad (1.8)$$

$$= \mathbb{E}[f(R \cos \theta, R \sin \theta)] \quad (1.9)$$

□

Remark 1.2.2.

General Gaussian vectors can be obtained by doing linear transformations on independent 1-dim Gaussian random variables, the coefficients of which are determined by the desired covariance matrix.

1.3 Reject Sampling

Consider picking m among n objects randomly where $m \nmid n$, for example choosing among 8 restaurants by tossing 3 coins, but 3 of them are unavailable. How can we build an uniform strategy of choice by still using these coins?

One way to do so is by resetting the problem after each pick until we land our choice upon an object among the wanted m ones.

The following results give a more general version of the reject sampling method.

Proposition 1.3.1.

X_n are i.i.d. random variables, A is a X -measurable event, and define $\tau = \inf \{n : X_n \in A\}$, then $X_\tau \sim X|_A$.

Proof. For any $g \in C_b(\mathbb{R})$,

$$\mathbb{E}[g(X_\tau)] = \sum_{n=1}^{\infty} \mathbb{E}[g(X_n) \mathbf{1}_{\tau=n}] \quad (1.10)$$

$$= \sum_{n=1}^{\infty} \mathbb{E}[g(X_n) \mathbf{1}_{X_n \in A}] \mathbb{P}(X_1, \dots, X_{n-1} \notin A) \quad (1.11)$$

$$= \sum_{n=1}^{\infty} (1-p)^{n-1} p \mathbb{E}[g(X) | X \in A] = \mathbb{E}[g(X) | X \in A] \quad (1.12)$$

where $p = \mathbb{P}(X_n \in A)$. □

Proposition 1.3.2.

If random variables X and Y have density functions $f(x)$ and $g(x)$, and $\exists$ constant C s.t. $Cg(x) \geq f(x)$. Sample a uniform distribution U over $[0, 1]$ s.t $U \perp Y$, then $\{Y | CUg(Y) < f(Y)\}$ samples the law of X . ■

Proof. For any $h \in C_b(\mathbb{R})$,

$$\mathbb{E}[h(Y) | f(Y) > CUg(Y)] = \frac{\int h(y) \frac{f(y)}{Cg(y)} g(y) dy}{\int \frac{f(y)}{Cg(y)} g(y) dy} \quad (1.13)$$

$$= \int h(y) f(y) dy \quad (1.14)$$

$$= \mathbb{E}[h(X)] \quad (1.15)$$

□

Example 1.3.3.

Given a coin without knowing p , p being the probability of tossing a head. Sample a uniform distribution among 3 options using this coin.

Toss the coin 3 times, do reject sampling on 6 of the 8 possible outcomes, and divide them into 3 groups:

1. +,-,-/-,+,+

2. -,+,-/+,-,+

3. -,+,+/-,+,+

The probability of each case is $p(1-p)^2 + p^2(1-p) = p(1-p)$, so the result of reject sampling is uniform among 3 options.

Lecture 2

MCMC: Monte-Carlo Markov Chains

After sampling 1-dimensional random variables with methods mentioned in the previous section, a natural question arises: what about random variables or probability measures on general spaces?

2.1 Motivating Examples

In this section we assume X is finite.

2.1.1 Boltzmann-Gibbs measure

Boltzmann-Gibbs measure $\pi(x) = \frac{1}{Z_\beta} \exp(-\beta H(x))$ is defined on a sample space X , where $H(x)$ is the Hamiltonian, also known as energy. $\beta = \frac{1}{T}$ is the reverse temperature, and $Z_\beta = \sum_{x \in X} \exp(-\beta H(x))$ is the partition function (Note that in the case $\beta = 0$, $\pi(x)$ is the uniform measure on X : every state has equal probability).

As $\beta \rightarrow +\infty$, $T \rightarrow 0$, $\pi(x)$ approximates the uniform measure on $\operatorname{argmin}_X H(x)$. This implies that if we want a point taking minimum value in some optimization problem, we can sample a Gibbs measure and take $\beta \rightarrow +\infty$.

2.1.2 Ising model

Set $X = \{-1, +1\}^V$, $G = (V, E)$ where $x \sim y$ if $\{x, y\} \in E$. Define the H function as:

$$H(\sigma) = - \sum_{x \sim y} \sigma_x \sigma_y, \quad \forall \sigma \in X, \sigma : V \rightarrow \{\pm 1\}$$

Observe that H attains minimum when Image of σ is 1 or -1 . In other words, measure μ prefers the states of the same sign. For $\mathbb{Z}^d, d \geq 2$, $\exists \beta_c$ such that the state is ordered when $\beta > \beta_c$ and disordered when $\beta < \beta_c$.

2.1.3 Spin glass: a generalization

Again set $X = \{\pm 1\}^V$, but in this case we have $H(\sigma) = -\sum_{x \sim y} J_{xy} \sigma_x \sigma_y$ with no more assumptions concerning the signs of the J_{xy} s.

A concrete case can be found when looking at the relationships between people. Label the edge $x \sim y$ with values J_{xy} to quantify how good the relationship between the two vertices is. Some specific examples are:

1. $V = \{1, 2, 3\}$, $J_{12} = 1$, $J_{13} = 1$, $J_{23} = -1$, $H(\sigma) = -x_1 x_2 - x_1 x_3 + x_2 x_3$. In both the cases $(x_1, x_2, x_3) = (1, 1, 1)$ and $(x_1, x_2, x_3) = (1, 1, -1)$, H attains minimum -1 . Thus in this case the values of x_2 and x_3 are not important.
2. $V = \{1, 2, 3\}$, $J_{12} = 1$, $J_{13} = 1$, $J_{23} = -100$, $H(\sigma) = -x_1 x_2 - x_1 x_3 + 100 x_2 x_3$. The minimum value is attained at $(x_1, x_2, x_3) = (1, 1, -1)$. In this case the value of x_1 is not important.

2.2 Backup Knowledge: Markov Chains

2.2.1 Basic concepts

Definition 2.2.1.

$(X_n)_{n \in \mathbb{N}}$ is a **Markov chain** with transition matrix **transition matrix** P_n if and only if

$$\mathbb{P}(X_n = x_n \mid X_0 = x_0, \dots, X_{n-1} = x_{n-1}) = \mathbb{P}(X_n = x_n \mid X_{n-1} = x_{n-1}) = P_n(x_{n-1}, x_n)$$

In other words, X_n satisfy the **Markov property**: the behaviour of X_{n+1} has nothing to do with X_i where $0 \leq i \leq n-1$.

In particular, if P_n does not depend on n , the Markov chain is **homogeneous**, otherwise it is **inhomogeneous**.

Definition 2.2.2.

The **generator** of a Markov Chain is $L = P - I$, where P is the transition matrix.

Proposition 2.2.3.

$$\mathbb{E}[f(X_n) \mid X_0 = x_0] = (P^n f)(x_0)$$

$$\mathbb{P}(X_n = \cdot \mid X_0 = x_0) = P^n(x_0, \cdot)$$

Proof. We prove the second statement by induction. For $n = 0$ and $n = 1$, the statement is

obviously true by definition. Suppose it's true for all $n \leq m$, then:

$$\begin{aligned}
 \mathbb{P}(X_{m+1} = \cdot | X_0 = x_0) &= \sum_x \mathbb{P}(X_{m+1} = \cdot, X_m = x | X_0 = x_0) \\
 &= \sum_x \mathbb{P}(X_{m+1} = \cdot | X_n = x, X_0 = x_0) \mathbb{P}(X_m = x | X_0 = x_0) \\
 &\stackrel{\text{Markov Property}}{=} \sum_x \mathbb{P}(X_{m+1} = \cdot | X_m = x) \mathbb{P}(X_n = x | X_0 = x_0) \\
 &= \sum_x P^m(x_0, x) P(x, \cdot) = P^{m+1}(x_0, \cdot)
 \end{aligned}$$

For the first statement, note that

$$\mathbb{E}[f(X_n) | X_0 = x_0] = \sum_x \mathbb{P}(X_n = x | X_0 = x_0) f(x) \quad (2.1)$$

$$= \sum_x P^n(x_0, x) f(x) = (P^n f)(x_0). \quad (2.2)$$

□

2.2.2 Invariant and stationary measures

Definition 2.2.4 (Invariant measure/stationary measure).

π is an **invariant measure** for (X_n) if and only if one of the following equivalent conditions holds:

1. $\forall y, \sum_x \pi(x) p(x, y) = \pi(y)$.
2. $\forall f, \mathbb{E}_\pi[f(X_1)] = \mathbb{E}_\pi[f(X_0)]$, where $\mathbb{E}_\pi$ denotes the (unconditional) expectation on $X_0 \sim \pi$.

Note that the first condition implies $X_n \sim \pi$.

Proof. $1 \Rightarrow 2$:

$$\mathbb{E}_\pi[f(X_1)] = \sum_{x,y} \mathbb{P}(X_0 = x, X_1 = y) f(y) = \sum_y \left(\sum_x \pi(x) p(x, y) \right) f(y) = \mathbb{E}_\pi[f(X_0)]$$

$2 \Rightarrow 1$:

$$0 = \mathbb{E}_\pi[f(X_1)] - \mathbb{E}_\pi[f(X_0)] = \sum_y f(y) \left(\sum_x \pi(x) p(x, y) - \pi(y) \right)$$

Take $f = \delta_y$, we obtain $\forall y, \sum_x \pi(x) p(x, y) = \pi(y)$. □

Definition 2.2.5.

(X_n) is **reversible w.r.t. measure** π if $\forall x, y \in V$,

$$\pi(x) p(x, y) = \pi(y) p(y, x)$$

Observe that reversibility implies invariance.

Remark 2.2.6.

Invariant measures are not necessarily unique. For a counter-example, consider a frog jumping between ponds, if the ponds are disconnected, there will be different invariant measures for distinct connected components.

Ergodicity

Definition 2.2.7.

An invariant measure π is **ergodic** if $\forall x_0 \in X, P_n(x_0, \cdot) \xrightarrow{n \rightarrow +\infty} \pi$.

Theorem 2.2.8.

Suppose X is finite, aperiodic and irreducible, then X_n is ergodic.

Theorem 2.2.9 (Birkhoff).

If π is ergodic, then $\frac{1}{n} \sum_{i=1}^n f(X_i) \xrightarrow{a.s.} \mathbb{E}_\pi[f]$

The proofs of the above theorems are taught in *Probability 2*, and therefore shall not be provided in this note.

2.3 MCMC algorithms

2.3.1 Birkhoff's algorithm

Suppose one is to find $\int_X f d\pi$, the following steps can be taken:

1. Construct a Markov chain with π as an ergodic measure.
2. Take $\widehat{M}_n = \frac{1}{n} \sum_{i=1}^n f(X_i)$, then $\widehat{M}_n \xrightarrow[n \rightarrow +\infty]{a.s.} \int_X f d\pi$.

An improvement to the second step is to sample Markov chains $X^{(1)}, \dots, X^{(M)}$ Markov chains and consider

$$\widehat{M_{N,M}}(f) = \frac{1}{MN} \sum_{i=1}^N \sum_{j=1}^M f(X_i^{(j)})$$

Two ways are given to construct the wanted Markov chain.

2.3.2 Glauber dynamics of Ising model

Take $\mu_\beta = \frac{1}{Z_\beta} \exp(\beta \sum_{x \sim y} \sigma_x \sigma_y)$ and repete the following operations:

1. Choose a vertex x uniformly randomly.
2. Flip the sign of x , i.e. $\sigma_x^{n+1} = -\sigma_x^n$, with probability

$$\frac{\exp(-\beta \sigma_x^n \sum_{y \sim x} \sigma_y^n)}{\exp(\beta \sum_{y \sim x} \sigma_y^n) + \exp(-\beta \sum_{y \sim x} \sigma_y^n)}$$

Proposition 2.3.1.

μ_β is ergodic measure of the Markov chain σ^n .

Proof. Write $\sigma_y^x = \begin{cases} -\sigma_x & \text{if } y = x \\ \sigma_y & \text{if } y \neq x \end{cases}$, it can be directly checked that: $\mu_\beta(\sigma) P(\sigma, \sigma^x) = \mu_\beta(\sigma^x) P(\sigma^x, \sigma)$.

Since μ_β is reversible, it is also a stationary method of the Markov chain. By definition, the chain is aperiodic, irreducible and finite, so μ_β is ergodic. $\square$

2.3.3 Metropolis-Hasting algorithm

Assume we are given a measure π and want to construct a Markov chain of which π is an ergodic measure. We have in hand a Markov chain X associated with transition probability matrix $p(x, y)$. How can we build the new MC?

Define another Markov chain with transition probability $\tilde{p}(x, y)$ satisfying:

$$\tilde{p}(x, y) = p(x, y) \min\left\{1, \frac{\pi(y) p(y, x)}{\pi(x) p(x, y)}\right\}$$

View $p(x, y)$ as the initiative to move from x to y , then the last term in this equation can be understood as the accepting probability, making $\tilde{p}(x, y)$ the actual probability to perform this move.

Observe that $\pi(x)\tilde{p}(x, y) = \pi(y)\tilde{p}(y, x)$. Then by a proof similar to that in Prop 1.2.10, the Markov chain generated by $\tilde{p}$ has ergodic measure π .

Note that this method can be used together with the rejection method.

2.4 Distance on measures

2.4.1 Total variation

Definition 2.4.1.

The **total variation distance** between two measure μ and ν is

$$d_{TV}(\mu, \nu) = \frac{1}{2} \sum_x |\mu(x) - \nu(x)| \quad (2.3)$$

$$= \sup_{A \subset X} (\mu(A) - \nu(A)) \quad (2.4)$$

$$= \sum_{\|f\|_\infty \leq 1} (\mu(f) - \nu(f)) \quad (2.5)$$

Example 2.4.2.

For $x \neq y$, $d_{TV}(\delta_x, \delta_y) = 1$.

2.4.2 Wasserstein distance

Definition 2.4.3.

For $1 \leq p < +\infty$, the **Wasserstein distance** between measures μ and ν is

$$W_p(\mu, \nu) = \inf_{(X,Y) \in \Gamma(\mu, \nu)} \mathbb{E}[|X - Y|^p]^{1/p}$$

where $(X, Y) \in \Gamma(\mu, \nu)$ denotes a coupling satisfying $X \sim \mu$, $Y \sim \nu$.

Example 2.4.4.

$$W_1(\delta_x, \delta_y) = |x - y|_1.$$

Proposition 2.4.5.

Let $\mathcal{X}$ be a Polish space, then there exists a coupling $(X, Y) \sim \Gamma(\mu, \nu)$ such that

$$\mathbb{E}[|X - Y|^p]^{1/p} = W_p(\mu, \nu)$$

2.4.3 More distances

Levy-Prokhorov, etc.

Lecture 3

Concentration and Functional inequalities

3.1 Confidence interval

Consider i.i.d. random variables X_i with mean $m = \mathbb{E}[X_i]$, set $\widehat{m}_N = \frac{1}{N} \sum_{i=1}^N X_i$. There are two ways to look at the problem of confidence interval of m using $\widehat{m}_N$'s:

1. Inequalities in the form of $\mathbb{P}(|\widehat{m}_n - m| \geq \epsilon) \leq \dots$
2. Statements in the form of $\widehat{m}_N \in [m - \delta(\epsilon), m + \delta(\epsilon)]$ has probability $1 - \epsilon$ of being true. Here $\delta(\epsilon)$ can be determined using CLT or Hoeffding Thm.

3.1.1 CLT

Assume $\sigma^2 = \mathbb{E}[X^2] < \infty$, then by Central Limit Theorem (CLT):

$$\mathbb{P}(\sqrt{N}|\widehat{m}_N - m| \leq \delta) \xrightarrow{N \rightarrow +\infty} \int_{-\delta}^{\delta} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

The disadvantages of this method are

1. It cannot be applied to fixed N , hence not available when one has limited data.
2. The probability $\mathbb{P}(|\widehat{m}_N - m| \leq \delta)$ is not contained directly in the statement above.

3.1.2 Hoeffding's Theorem

Theorem 3.1.1.

Suppose X_i are independent random variables with $X_i \in [a_i, b_i]$, and let $S_n = \sum_{i=1}^n X_i$. Then

$$\mathbb{P}(S_n - \mathbb{E}[S_n] \geq \epsilon) \leq \exp\left(-\frac{2\epsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right)$$

Proof. **Step 1** Markov property + Laplace transform

$$\mathbb{P}(X \geq x) \leq e^{-\lambda x} \mathbb{E}[\exp(\lambda x)] = e^{\varphi_X(\lambda) - \lambda x}$$

where $\varphi_X(\lambda) = \log \mathbb{E}[e^{\lambda X}]$ is defined by Laplace transform.

Step 2 Properties about Laplace transform function

Assume X is centered, $\mathbb{E}[X] = 0$, and X is bounded in $[a, b]$. Then $\lambda \rightarrow \varphi_X(\lambda)$ is well-defined and analytic, and we have

$$\varphi_X(\lambda) = \varphi_X(0) + \int_0^\lambda \varphi'_X(t) dt$$

Since $\varphi_X(0) = 0$, $\varphi'_X(0) = 0$, and

$$\varphi'_X(\lambda) = \frac{\mathbb{E}[X e^{\lambda X}]}{\mathbb{E}[e^{\lambda X}]}$$

$$\varphi''_X(\lambda) = \frac{\mathbb{E}[X^2 e^{\lambda X}]}{\mathbb{E}[e^{\lambda X}]} - \left(\frac{\mathbb{E}[X e^{\lambda X}]}{\mathbb{E}[e^{\lambda X}]} \right)^2$$

Then,

$$\varphi_X(\lambda) = \int_0^\lambda \int_0^t \varphi''_X(s) ds$$

Do measure transformation:

$$\frac{d\mathbb{P}^\lambda}{d\mathbb{P}} = \frac{e^{\lambda X}}{\mathbb{E}[e^{\lambda X}]}$$

Then

$$\mathbb{E}^\lambda[f(X)] = \mathbb{E}\left[\frac{e^{\lambda X}}{\mathbb{E}[e^{\lambda X}]} f(X)\right]$$

So

$$\varphi''_X(\lambda) = \text{Var}^\lambda[X]$$

$$\varphi'_X(\lambda) = \mathbb{E}^\lambda[X]$$

Step 3 Inequalities

Since $X \in [a, b]$, so $\text{Var}_Q[X] \leq \frac{(b-a)^2}{4}$ for any measure Q . Indeed,

$$\text{Var}_Q[X] = \min_{x \in \mathbb{R}} \mathbb{E}_Q[(X - x)^2] \leq \mathbb{E}_Q\left[\left(X - \frac{a+b}{2}\right)^2\right] \leq \frac{(b-a)^2}{4}$$

So

$$\varphi_X(\lambda) \leq \frac{(b-a)^2 \lambda^2}{8} + \lambda \mathbb{E}[X]$$

Since

$$\mathbb{P}(X - \mathbb{E}[X] \geq x) \leq \exp(-\lambda x + \varphi_X(\lambda) - \mathbb{E}[X] \lambda)$$

So

$$\begin{aligned}
\mathbb{P}(S_n - \mathbb{E}[S_n] \geq x) &\leq \exp(\varphi_{S_n}(\lambda) - \lambda x - \mathbb{E}[S_n] \lambda) \\
&\leq \exp\left(\sum_{i=1}^n \varphi_{X_i}(\lambda) - \lambda x - \sum_{i=1}^n \mathbb{E}[X_i] \lambda\right) \\
&\leq \exp\left(\frac{\lambda^2}{8} \sum_{i=1}^n (b_i - a_i)^2 - \lambda x\right)
\end{aligned}$$

Take λ such that the result is minimum, i.e. $\lambda = \frac{4x}{\sum_{i=1}^n (b_i - a_i)^2}$, then we have

$$\mathbb{P}(S_n - \mathbb{E}[S_n] \geq x) \leq \exp\left(-\frac{2x^2}{\sum_{i=1}^n (b_i - a_i)^2}\right)$$

□

3.2 Efron-Stein inequality and McDiarmid Theorem

First we prove the Efron-Stein inequality.

Theorem 3.2.1 (Efron-Stein).

Suppose X_i are independent random variables, X'_i be an independent copy of X_i , i.e. $X'_i \stackrel{d}{=} X_i$, and $X'_i \perp X_i$. Then

$$\text{Var}[f(X_1, \dots, X_n)] \leq \sum_{i=1}^n \mathbb{E}[(f(X_1, \dots, X_n) - f(X'_1, \dots, X'_n))^2]$$

Proof. Denote $Z = f(X_1, \dots, X_n)$ and $\mathcal{F}_i = \sigma(X_1, \dots, X_i)$, then

$$\text{Var}[Z] = \mathbb{E}[(Z - \mathbb{E}[Z | \mathcal{F}_0])^2] = \sum_{i=1}^n \mathbb{E}[(\mathbb{E}[Z | \mathcal{F}_i] - \mathbb{E}[Z | \mathcal{F}_{i-1}])^2]$$

This process is called the *orthogonal decomposition*. Now we deal with each $\mathbb{E}[(\mathbb{E}[Z | \mathcal{F}_i] - \mathbb{E}[Z | \mathcal{F}_{i-1}])^2]$: For any g , know that

$$\mathbb{E}[(g - \mathbb{E}_{X_i}[g])^2] = \mathbb{E}[\mathbb{E}_{X_i}[(g - \mathbb{E}_{X_i}[g])^2]]$$

$$\mathbb{E}_{X_i}[g] = \mathbb{E}_{X'_i}[g(X_1, \dots, X'_i)]$$

So

$$\begin{aligned}
\mathbb{E}_{X_i}[(g(\dots, X_i) - \mathbb{E}_{X'_i}[g(\dots, X'_i)])^2] &= \mathbb{E}_{X_i}[(\mathbb{E}_{X'_i}[g(\dots, X_i) - g(\dots, X'_i)])^2] \\
&\leq \mathbb{E}_{X_i}[\mathbb{E}_{X'_i}[(g(\dots, X_i) - g(\dots, X'_i))^2]]
\end{aligned}$$

Let $g(X_1, \dots, X_i) = \mathbb{E}_{X_{i+1}, \dots, X_n} [f]$, then

$$\begin{aligned} \mathbb{E}[(\mathbb{E}[Z|\mathcal{F}_i] - \mathbb{E}[Z|\mathcal{F}_{i-1}])^2] &\leq \mathbb{E}_{X_i}[\mathbb{E}_{X'_i}[(\mathbb{E}_{X_{i+1}, \dots, X_n}[f(X_i) - f(X'_i)])^2]] \\ &\stackrel{Jensen}{\leq} \mathbb{E}[(f(X_i) - f(X'_i))^2] \end{aligned}$$

Hence the inequality is proved. $\square$

This inequality can be used to prove the McDiarmid Theorem, as stated below.

Theorem 3.2.2 (McDiarmid).

For function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ satisfying $|f(x) - f(y)| \leq \sum_{i=1}^n c_i \mathbf{1}_{x_i \neq y_i}$, take independent random variables $X_1, \dots, X_n$ and define $Z = f(X_1, \dots, X_n)$. Then

$$\mathbb{P}(Z - \mathbb{E}[Z] \geq \epsilon) \leq \exp\left(-\frac{2\epsilon^2}{\sum_{i=1}^n c_i^2}\right)$$

Proof. Step 1 Use Markov property and Laplace function

We can easily see that

$$\mathbb{P}(Z - \mathbb{E}[Z] \geq x) \leq \exp(\varphi_Z(\lambda) - \lambda x - \mathbb{E}[Z]\lambda)$$

where φ_Z is defined as previously in this section.

Step 2 Log-Laplace

$$\begin{aligned} \varphi_Z(\lambda) &\leq \int_0^\lambda \int_0^t \varphi_Z''(s) ds + \lambda \mathbb{E}[Z] \\ \varphi_Z''(s) &= \text{Var}^s[Z] \leq \sum_{i=1}^n \mathbb{E}^s[(f(X_i) - f(X'_i))^2] \leq \sum_{i=1}^n c_i^2 \end{aligned}$$

$\square$

Corollary 3.2.3.

Suppose random variable $X \stackrel{d}{\sim} \mathcal{N}(0, 1)$, then

1. $\text{Var}[f(X)] \leq \mathbb{E}[f'(X)]$ for any $f \in C^1(\mathbb{R})$.
2. $\text{Var}[f(X)] \leq \mathbb{E}[|\nabla f(X)|^2]$ for any $f \in C^1(\mathbb{R}^n)$.

Proof. Firstly, (1) $\rightarrow$ (2) is obvious using the Efron-Stein inequality. So it suffices to prove (1).

Construct i.i.d. Y_i s.t. $\mathbb{P}(Y = 1) = \mathbb{P}(Y = -1) = \frac{1}{2}$. By CLT, for any $f \in C^1 \cap C_b$,

$$\text{Var}[f(X)] = \lim_{n \rightarrow \infty} \text{Var}\left[f\left(\frac{\sum_{i=1}^n Y_i}{\sqrt{n}}\right)\right]$$

Denote $\frac{\sum_{i=1}^n Y_i}{\sqrt{n}} = X^n$, then

$$\begin{aligned} \text{Var} \left[f \left(\frac{\sum_{i=1}^n Y_i}{\sqrt{n}} \right) \right] &\leq \sum_{k=1}^n \mathbb{E} \left[\left(f \left(\frac{\sum_{i=1}^n Y_i}{\sqrt{n}} \right) - f \left(\frac{\sum_{i \neq k} Y_i + Y'_k}{\sqrt{n}} \right) \right)^2 \right] \\ &\leq \sum_{k=1}^n \mathbb{E} \left[\left(f'(X^n) \cdot \frac{1}{\sqrt{n}} \right)^2 \right] \\ &= \mathbb{E} [f'(X^n)^2] \xrightarrow{n \rightarrow \infty} \mathbb{E} [f'(X)^2] \end{aligned}$$

□

3.3 Supplement: Spectral properties of Markov Chains

Research on spectral properties of Markov chains focuses on the spectrum of the transition matrices, especially the extremal eigenvalues, which determines characteristics of the chain such as mixing time, convergence rate and bottleneck behaviour.

Assume P is the transition matrix of an irreducible, aperiodic, reversible Markov chain on finite state space Ω with stationary distribution π . Suppose λ_i are the eigenvalues of P , and f_i is a set of eigen-functions orthonormal under inner product $(f, g) = \sum_{x \in \Omega} f(x) g(x) \pi(x)$. Obviously the eigenvalues of P satisfy

$$1 = \lambda_1 > \lambda_2 \geq \dots \geq \lambda_n \geq -1$$

Definition 3.3.1. 1. The **spectral gap** of an irreducible, aperiodic, reversible Markov chain with eigenvalues λ_i is $1 - \lambda_2$.

2. The **absolute spectral gap** is $1 - \max \{ \lambda_2, |\lambda_n| \}$.

3. $\lambda_{\pm} = \max \{ |\lambda_i|, i \neq 1 \}$.

Define $d(t) = \|p_n(x, \cdot) - \pi\|_{TV}$, where $p_n(x, y)$ is the probability of moving from state x to y after n steps, and define $\nu_x^{(i)} = f_i(x)$. Then

$$p_n(x, y) = \delta_{x,y} P^n = \sum_x \nu_x P^n(y) = \pi(y) + \sum_i \nu_x^{(i)} \nu_y^{(i)} \pi(y) \lambda_i^n$$

Theorem 3.3.2.

Let $\pi_{\min} = \min_{x \in \Omega} \pi(x)$, then

$$d(t) \leq \frac{1}{2\sqrt{\pi_{\min}}} \lambda_{\pm}^t$$

Now we give two theorems concerning mixing time $t_{\text{mix}} = \inf \{ t : \sup_X \|p_t(x, \cdot) - \pi\|_{TV} \leq \epsilon \}$. ■

Theorem 3.3.3 (Ding-Peres, 2013).

For any graph G of n vertices with edge weights J ,

$$\inf_{G, J} t_{\text{mix}}^+(G, J) \geq \left(\frac{1}{4} + o(1)\right) n \log n$$

Theorem 3.3.4 (Lubetzky-Sly, 2009).

Cutoff happens for Ising model on $\mathbb{Z}^d / n\mathbb{Z}$, i.e. when the temperature is above a critical value, mixing time is strictly $O(n \log n)$, and the spectral gap is strictly positive. In short words, the particles exhibit strong mixing behaviour.

Remark 3.3.5.

Before Lubetzky and Sly, mathematicians were only able to prove that $d = 2$ case has a slow growth of $O(\log \log n)$.

3.4 Correction of preceding proofs

Hoeffding's Inequality

Let X_i be independent random variables such that $X_i \in [a_i, b_i]$. Let $S_n = \sum_{i=1}^n X_i$.

$$\Rightarrow \mathbb{P}(|S_n - \mathbb{E}[S_n]| \geq \epsilon) \leq \exp\left(-\frac{2\epsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right)$$

Proof (Centralization): Without loss of generality (WMA), assume $\mathbb{E}[X_i] = 0$, which implies $\mathbb{E}[S_n] = 0$.

$$\mathbb{P}(S_n - \mathbb{E}[S_n] \geq \epsilon) = \mathbb{P}(S_n \geq \epsilon) \leq \mathbb{P}(e^{\lambda S_n} \geq e^{\lambda \epsilon}) \leq e^{-\lambda \epsilon} \mathbb{E}[e^{\lambda S_n}] = e^{-\lambda \epsilon} \prod_{i=1}^n \mathbb{E}[e^{\lambda X_i}] \quad \otimes$$

Consider the individual terms: $\mathbb{E}[e^{\lambda X_i}] = e^{\log \mathbb{E}[e^{\lambda X_i}]}$. Let $\varphi_{X_i}(\lambda) = \log \mathbb{E}[e^{\lambda X_i}]$. Since $\varphi_{X_i}(0) = 0$, using a Taylor expansion (or double integration form):

$$\varphi_{X_i}(\lambda) = \int_0^\lambda \varphi'_{X_i}(t) dt + \varphi_{X_i}(0) = \int_0^\lambda \varphi'_{X_i}(t) dt, \quad \text{where } \varphi'_{X_i}(t) = \frac{\mathbb{E}[X_i e^{tX_i}]}{\mathbb{E}[e^{tX_i}]}$$

$$\varphi_{X_i}(\lambda) = \int_0^\lambda \int_0^t \varphi''_{X_i}(s) ds dt = \int_0^\lambda \int_0^t \text{Var}^s(X_i) ds dt$$

From the variance bound derived below:

$$\leq \frac{1}{2} \lambda^2 \cdot \frac{(b_i - a_i)^2}{4} = \frac{\lambda^2 (b_i - a_i)^2}{8}$$

Substituting this back into the original expression $\otimes$ yields:

$$\Rightarrow \otimes \leq e^{-\lambda\epsilon + \frac{\lambda^2}{8} \sum_{i=1}^n (b_i - a_i)^2}$$

Lemmas for Efron-Stein Inequality

$$\varphi_X''(s) = \frac{\mathbb{E}[X^2 e^{sX}]}{\mathbb{E}[e^{sX}]} - \left(\frac{\mathbb{E}[X e^{sX}]}{\mathbb{E}[e^{sX}]} \right)^2$$

Introducing the tilted measure (Change of measure): $\frac{d\mu_s}{d\mu} = \frac{e^{sX}}{\mathbb{E}[e^{sX}]}$, we have:

$$\text{Var}^s(X) = \min_{c \in \mathbb{R}} \mathbb{E}^s[(X - c)^2] \leq \mathbb{E}^s \left[\left(X - \frac{a_i + b_i}{2} \right)^2 \right] \leq \frac{(b_i - a_i)^2}{4}$$

Efron-Stein Inequality

Let (X_i) be independent random variables. Let (X'_i) be an independent copy of (X_i) , meaning $X'_i \sim X_i$ and $(X'_i) \perp (X_i)$.

$$\text{Var}(f(X_1, \dots, X_n)) \leq \frac{1}{2} \sum_{i=1}^n \mathbb{E} \left[(f(X_1, \dots, X_i, \dots, X_n) - f(X_1, \dots, X'_i, \dots, X_n))^2 \right]$$

Using Martingale Differences Decomposition: Let $g = f(X_1, \dots, X_n)$ [cite: 20], and denote $\mathbb{E}_{X_i}[g]$ as the integration with respect to X_i .

$$\mathbb{E}[(\mathbb{E}[Z|\mathcal{F}_{i+1}] - \mathbb{E}[Z|\mathcal{F}_i])^2] = \mathbb{E}[(\mathbb{E}_{X_i}[g] - g)^2]$$

$$\begin{aligned} \mathbb{E}_{X_i} [(\mathbb{E}_{X_i}[g] - g)^2] &= \mathbb{E}_{X_i} \left[(\mathbb{E}_{X'_i}[g(\dots, X'_i)] - g(\dots, X_i))^2 \right] \\ &= \mathbb{E}_{X_i} \left[(\mathbb{E}_{X'_i}[g(\dots, X'_i)] - g(\dots, X_i))^2 \right] \end{aligned}$$

By Jensen's Inequality:

$$\leq \mathbb{E}_{X_i} \mathbb{E}_{X'_i} [(g(\dots, X'_i) - g(\dots, X_i))^2]$$

Due to symmetry, this equals:

$$= \frac{1}{2} \mathbb{E}_{X_i} \mathbb{E}_{X'_i} [(g(\dots, X_i) - g(\dots, X'_i))^2]$$

Taking the total expectation on both sides and combining across the filtration $\mathcal{F}_i$:

$$\Rightarrow \mathbb{E}[(\mathbb{E}[Z|\mathcal{F}_{i+1}] - \mathbb{E}[Z|\mathcal{F}_i])^2] \leq \frac{1}{2} \mathbb{E} [(f(X) - f(X^{(i)}))^2]$$

McDiarmid's Inequality (Bounded Differences Inequality)

Let $f : \mathcal{X}^n \rightarrow \mathbb{R}$ be a function satisfying the bounded differences condition:

$$|f(x) - f(y)| \leq \sum_{i=1}^n c_i \mathbb{I}_{\{x_i \neq y_i\}}$$

Let $X_1, \dots, X_n$ be independent random variables, and let $Z = f(X_1, \dots, X_n)$.

$$\Rightarrow \mathbb{P}(Z - \mathbb{E}[Z] \geq \epsilon) \leq \exp\left(-\frac{2\epsilon^2}{\sum_{i=1}^n c_i^2}\right) \quad [\text{cite: 32}]$$

Proof Proof (Centralization): Without loss of generality, assume $\mathbb{E}[Z] = 0$.

$$\mathbb{P}(Z - \mathbb{E}[Z] \geq \epsilon) \leq e^{-\lambda\epsilon + \varphi_Z(\lambda)}$$

where $\varphi_Z(\lambda) = \int_0^\lambda \int_0^t \text{Var}^s(Z) ds dt$ (as with Hoeffding). By martingale decomposition and conditional variance bounding:

$$\begin{aligned} \text{Var}^s(Z) &= \sum_{i=1}^n \mathbb{E}^s \left[(\mathbb{E}^s[Z | X_1, \dots, X_i] - \mathbb{E}^s[Z | X_1, \dots, X_{i-1}])^2 \right] \\ &= \sum_{i=1}^n \mathbb{E}^s \left[(\mathbb{E}^s[Z - \mathbb{E}^s[Z | X_1, \dots, X_i] | X_1, \dots, X_{i-1}])^2 \right] \\ &\leq \sum_{i=1}^n c_i^2 \end{aligned}$$

Lecture 4

Extreme Value Theory and Importance Sampling

4.1 Importance sampling

We wish to estimate $\mathbb{E}[f(X)]$ for some random variable X .

A natural idea is to sample $X_1, X_2, \dots, X_M$, from which we obtain an estimator

$$\hat{f} = \frac{1}{M} \sum_{i=1}^M f(X_i)$$

Note that $\mathbb{E}[\hat{f}] = \mathbb{E}[f]$, yet $\text{Var}[\hat{f}] = \frac{1}{M} \text{Var}[f] \rightarrow 0$ as $M \rightarrow \infty$.

Example 4.1.1 (Rare event).

Let random variable $X \sim \mathcal{N}(0, 1)$, and $f(X) = \mathbb{1}_{\{X \geq a\}}$. Take $a = 2/3$, denote $p = \mathbb{E}[f]$ and $\text{Var}[f] = p(1-p)$. Note that $p \approx 10^{-3}$, so $X \geq a$ is a rare event.

Observe that if $\text{Var}[\hat{f}]$ is relatively large and $\mathbb{E}[f]$ small, then the noise makes it difficult to estimate the mean. Thus we wish for

$$\frac{\sqrt{\text{Var}[\hat{f}]}}{\mathbb{E}[\hat{f}]} \ll 1$$

In the case of the rare event,

$$\frac{\sqrt{\frac{1}{M} p(1-p)}}{p} \approx r \implies M \approx \frac{1}{pr^2}$$

so for $p \approx 10^{-3}$ and ideal $r \approx 10^{-1}$ we need $M \approx 10^5$ to avoid too much noise.

In general, we wish to reduce $\text{Var}[\hat{f}]$. To do this, we introduce a method of building new

measures.

Definition 4.1.2 (Radon-Nikodym derivative).

Let $Q \ll P$ be two probability measures, define

$$\frac{dQ}{dP} = L$$

where L is the **Radon-Nikodym derivative of Q with respect to P** .

Then for any event A and any random variable $X \in P$,

$$\mathbb{E}_Q[A] = \mathbb{E}_P[L \cdot 1_{\{A\}}]$$

$$\mathbb{E}_Q[f(X)] = \mathbb{E}_P[L \cdot f(X)], \quad \mathbb{E}_P[f(X)] = \mathbb{E}_Q[L^{-1}f(X)]$$

Now we present the **importance sampling algorithm** using the Radon-Nikodym derivative.

1. Sample X under probability measure Q .

2. Calculate $\hat{f} = \frac{1}{M} \sum_{i=1}^M f(X_i) L^{-1}(X_i)$.

Example 4.1.3 (Importance sampling for rare event of Gaussian).

Define $f = 1_{x \geq a}$ for $a > 0$.

1. Sample $X \sim \mathcal{N}(0, 1)$ and $Y \sim \mathcal{N}(a, 1)$.

2. $\hat{f} = \frac{1}{M} \sum_{i=1}^M f(Y_i) L^{-1}(Y_i)$.

We hope to find a measure derivative L such that $\mathbb{E}[f(X)] = \mathbb{E}[\hat{f}(Y)]$.

It would be fine to calculate first the Radon-Nikodym derivative and then invert it, but there is an easier way to find L . Note that

$$LHS = \int_{\mathbb{R}} 1_{\{X \geq a\}} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

$$RHS = \int_{\mathbb{R}} 1_{\{X \geq a\}} \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-a)^2}{2}} L^{-1}(x) dx$$

By $LHS = RHS$, we get

$$L^{-1}(x) \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-a)^2}{2}} = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

$$L^{-1}(x) = e^{-ax + \frac{a^2}{2}}$$

Hence we obtain $\hat{f} = \frac{1}{M} \sum_{i=1}^M f(Y_i) e^{-aY_i + \frac{a^2}{2}}$.

Remark 4.1.4.

Why do we calculate the expectation under a different measure? Because when $a > 0$, the result of f can be frequently 0, which is invalid data for calculating $\mathbb{E}[f(X)]$, but after a transformation of measures, $\hat{f}$ gives better results and more useful data. Then why must Y be $\mathcal{N}(a, 1)$? Actually, if we take $Y \sim \mathcal{N}(b, 1)$ with $b > a$, f would be rarely 0, which also causes abnormality in the resulting data.

4.2 Extreme value theory

Theorem 4.2.1 (Gumbel-Frechet-Weibull).

Let (X_i) be i.i.d random variables. $M_n = \max_{1 \leq i \leq n} X_i$. If there exist some a_n, b_n and random variable Y such that

$$\frac{M_n - b_n}{a_n} \xrightarrow{n \rightarrow +\infty} Y$$

then Y can only follow one of the three distributions: **Gumbel, Frechet and Weibull**.

We now give explicit definitions of the three distributions mentioned above. Let $G(x) = \mathbb{P}(Y \leq x)$.

4.2.1 Gumbel: light tail

$$G(x) = \exp(-e^{-x})$$

This applies to the random variables X with fast decaying tails, such as the Gaussian and the exponential random variables.

Example 4.2.2 (exponential).

Let $\mathbb{P}(X \leq x) = 1 - e^{-x}$, $b_n = \log n$, $a_n = 1$.

$$\begin{aligned} \mathbb{P}(M_n - \log n \leq x) &= \mathbb{P}(M_n \leq \log(n) + x) \\ &= \mathbb{P}(\cap_{i=1}^n \{X_i \leq \log n + x\}) \\ &= \prod_{i=1}^n \mathbb{P}(X_i \leq \log n + x) \\ &= (1 - \exp(-(\log n + x)))^n \\ &= (1 - \frac{e^{-x}}{n})^n \xrightarrow{n \rightarrow +\infty} \exp(-e^{-x}) \end{aligned}$$

Example 4.2.3 (Gaussian (exercise)).

Let

$$\begin{aligned} b_n &= \sqrt{2 \log n} - \frac{\log \log n + \log 4\pi}{2\sqrt{2 \log n}} \\ a_n &= (2 \log(n))^{-1/2} \end{aligned}$$

Here we only provide some intuition. Recall that

$$\int_x^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy \approx \frac{1}{x} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

and that

$$M_n \approx x \implies \int_x^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy = \frac{1}{n}$$

Roughly OMIT the $\frac{1}{x}$ term in the approximation of the tail probabilities. Then $\frac{1}{\sqrt{2\pi}} e^{-\frac{\tilde{x}^2}{2}} = \frac{1}{n}$ gives

$$\tilde{x} = \sqrt{2 \log n}$$

Writing $x = \tilde{x} + \delta$, we simplify to get $\frac{1}{\tilde{x}} \frac{1}{\sqrt{2\pi}} e^{-\tilde{x}\delta} = 1$

$$\begin{aligned} \delta &= \frac{-\log(\tilde{x} \sqrt{2\pi})}{\tilde{x}} \\ &\approx \frac{-\log(\sqrt{4\pi \log n})}{\sqrt{2 \log n}} \\ &= \frac{-\log(4\pi) - \log \log n}{2\sqrt{2 \log n}} \end{aligned}$$

4.2.2 Frechet: heavy tail

$$G(x) = \begin{cases} 0 & x \leq 0 \\ \exp(-x^\alpha) & x > 0 \end{cases}, \alpha > 0$$

This applies, in the theorem above, to heavy distributions, with $\mathbb{P}(X > x) \sim x^{-\alpha}$. In real life, such distributions may be used to model assets.

Example 4.2.4 (Pareto distribution).

Let $\mathbb{P}(X > x) = x^{-\alpha}$, $a_n = n^{\frac{1}{\alpha}}$, $b_n = 0$.

$$\begin{aligned} \mathbb{P}(M_n \leq n^{\frac{1}{\alpha}} x) &= \mathbb{P}(\cap_{i=1}^n \{X_i \leq n^{\frac{1}{\alpha}} x\}) \\ &= \mathbb{P}(X_i \leq n^{\frac{1}{\alpha}} x)^n = (1 - \frac{x^{-\alpha}}{n})^n \xrightarrow{n \rightarrow +\infty} \exp(-x^{-\alpha}) \end{aligned}$$

4.2.3 Weibull: null tail

$$G(x) = \begin{cases} \exp(-(-x)^\alpha) & x \leq 0 \\ 1 & x > 0 \end{cases}, \alpha > 0$$

This applies to bounded random variables, an interesting example of which is the distribution of examination scores.

Example 4.2.5 (Uniform distribution).

Let $X \sim \text{Unif}[0, 1]$, $a_n = \frac{1}{n}$, $b_n = 1$. For $x < 0$,

$$\begin{aligned} \mathbb{P}\left(\frac{M_n - 1}{\frac{1}{n}} \leq x\right) &= \mathbb{P}\left(M_n \leq 1 + \frac{x}{n}\right) \\ &= \mathbb{P}\left(X \leq 1 + \frac{1}{n}x\right)^n \xrightarrow{n \rightarrow +\infty} e^x = \exp(-(-x)^1) \end{aligned}$$

Example 4.2.6 (Beta distribution).

Define $B(\alpha, \beta) = \text{Beta}(\alpha, \beta) = \int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx$, where $\alpha, \beta > 0$. The distribution has density

$$\frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1} \mathbb{1}_{[0,1]}$$

Consider $X \sim \text{Beta}(1, \alpha)$ and $B(1, \alpha) = \frac{1}{\alpha}$, then

$$\mathbb{P}(X \leq x) = \int_0^x \alpha (1-y)^{\alpha-1} dy = 1 - (1-x)^\alpha$$

In this case, let $b_n = 1$, $a_n = n^{-\frac{1}{\alpha}}$.

$$\begin{aligned} \mathbb{P}\left(\frac{M_n - b_n}{a_n} \leq x\right) &= \mathbb{P}(M_n \leq 1 + n^{-\frac{1}{\alpha}}x) \\ &= \mathbb{P}(X \leq 1 + n^{-\frac{1}{\alpha}}x)^n = (1 - (-n^{\frac{1}{\alpha}}x)^\alpha)^n = \left(1 - \frac{1}{n}(-x)^\alpha\right)^n \\ &\xrightarrow{n \rightarrow +\infty} \exp(-(-x)^\alpha) \end{aligned}$$

4.3 Extreme value via Poisson point process

Definition 4.3.1.

$N \in M_\delta(S)$ is a **Poisson point process (PPP)** with density ν if and only if

1. $\forall A \in \mathcal{B}(S)$ (Borel sets), $N(A) \sim \text{Poisson}(\nu(A))$.
2. For $A_1, A_2, \dots, A_n \in \mathcal{B}(S)$ disjoint, $N(A_1), \dots, N(A_n)$ are independent.

Theorem 4.3.2.

For the three different types of tails distributions:

(1) Gumbel

Suppose (X_i) are i.i.d Gaussian r.v., then

$$\Lambda_n = \sum_{i=1}^n \delta_{\frac{X_i - b_n}{a_n}} \xrightarrow{n \rightarrow +\infty} \text{PPP with density } e^{-x} dx$$

The intuition is as follows: $\mathbb{P}(\frac{M_n - b_n}{a_n} \leq x) \approx \mathbb{P}(N[(x, +\infty)] = 0) = e^{-e^{-x}} = \exp(-e^{-x})$ where $N[(x, +\infty)] = Y \sim \text{Poisson}(e^{-x})$.

(2) *Frechet*

Let $\mathbb{P}(X \geq x) \sim \frac{c}{x^\alpha}$, then

$$\Lambda_n = \sum_{i=1}^n \delta_{n^{-\frac{1}{\alpha}} X_i} \xrightarrow{n \rightarrow +\infty} \text{PPP with density } \frac{c \alpha}{x^{\alpha+1}} dx$$

(3) *Weibull*

For $x < 0$, let $\mathbb{P}(X \geq x) \sim c|x|^\alpha$, then

$$\Lambda_n = \sum_{i=1}^n \delta_{n^{\frac{1}{\alpha}} X_i} \xrightarrow{n \rightarrow +\infty} \text{PPP with density } c \alpha |x|^{\alpha-1} dx$$

Lecture 5

Universality and condensation

In industrial experiments, engineering and in real life, we often encounter cases where some global estimator is considerably distant from that expected, thanks to only a certain few outliers. In other words, **global deviation of average data can be traced back to global local anomalies with a probability not so small**. The following theorem is an example of such phenomenon.

Theorem 5.0.1.

Let X_i be i.i.d. Poisson random variables, hence $\mathbb{P}(X_i > x) \sim \frac{1}{x^\alpha}$, where $\alpha \in (0, 1)$. Let $S_n = \sum_{i=1}^n X_i$, $M_n = \max_{1 \leq i \leq n} X_i$, then $\exists c \in (0, 1)$ s.t.

$$\liminf_{n \rightarrow \infty} \mathbb{P}\left(\frac{M_n}{S_n} > c\right) > c$$

In the rest of this lecture we work to prove this theorem.

5.1 Warm-up examples

Recall that CLT shows an asymptotic Gaussian behaviour, i.e. for i.i.d. X_i with mean m and variance σ^2 , we have

$$\frac{\sum_{i=1}^n (X_i - m)}{\sigma \sqrt{n}} \Rightarrow \mathcal{N}(0, 1)$$

A natural question arises: What happens if X_i doesn't have finite second moment?

Theorem 5.1.1.

Suppose i.i.d. random variables X_i satisfy $\mathbb{P}(X_i > x) = \mathbb{P}(X_i < -x) = \frac{x^{-\alpha}}{2}$, for a fixed $\alpha \in (0, 2)$ and any $x > 1$, then

$$\frac{S_n}{n^{\frac{1}{\alpha}}} \Rightarrow Y$$

where the characteristic function of Y is $\Phi_Y(t) = \exp(-c|t|^\alpha)$

$$\text{Proof. } \mathbb{E}[|X_i|^2] = 2 \int_1^{+\infty} x^{-\alpha} \cdot 2x \, dx = \begin{cases} +\infty & \alpha \in (0, 2) \Rightarrow \text{CLT stable} \\ < +\infty & \alpha \in (2, +\infty) \Rightarrow \text{CLT Gaussian} \\ +\infty & \alpha = 2 \Rightarrow \sqrt{n \log n} \text{ Gaussian} \end{cases}$$

The last case comes from a conclusion in **Probablity 1**: when $\alpha = 2$, $\frac{S_n}{\sqrt{n \log n}} \Rightarrow \mathcal{N}(0, \sigma^2)$
Now it suffices to prove that for $\alpha \in (0, 2)$,

$$\Phi_{S_n/n^{\frac{1}{\alpha}}}(t) = \Phi_{S_n}(n^{-\frac{1}{\alpha}} t) = \Phi_{X_i}(n^{-\frac{1}{\alpha}} t)^n \implies \Phi_Y(t)$$

Note that

$$\begin{aligned} \Phi_{X_i}(\theta) &= \int_1^{+\infty} \frac{\cos \theta x}{x^{1+\alpha}} \, dx = \theta^\alpha \int_\theta^{+\infty} \frac{\cos y}{y^{1+\alpha}} \, dy \\ 1 - \Phi_{X_i}(\theta) &= \theta^\alpha \int_\theta^{+\infty} \frac{1 - \cos y}{y^{1+\alpha}} \, dy \end{aligned}$$

Since $1 - \cos y \sim \frac{1}{2} y^2$ when y is near 0, so

$$\int_\epsilon^1 \frac{1 - \cos y}{y^{1+\alpha}} \, dy \sim y^{2-\alpha} \Big|_\epsilon^1 < 1$$

Thus $\int_\theta^{+\infty} \frac{\cos y}{y^{1+\alpha}} \, dy$ converges to a constant C as $\theta \rightarrow 0$. So $\exists$ constant c , such that

$$\Phi_{S_n/n^{\frac{1}{\alpha}}}(t) \sim (1 - C(n^{-\frac{1}{\alpha}} t)^\alpha)^n \xrightarrow{n \rightarrow +\infty} \exp(-ct^\alpha)$$

□

5.2 A deeper version of CLT

Theorem 5.2.1.

X_i are i.i.f random variables satisfying conitions:

- (Symmetry/Asymmetry) $\lim_{x \rightarrow +\infty} \mathbb{P}(X_i > x) / \mathbb{P}(X_i > |x|) = \theta \in [0, 1]$
- $\mathbb{P}(|X| > x) = x^{-\alpha} L(x)$, where $\alpha \in (0, 2)$ and L is a **slow-vary function**: for fixed $t > 1$, $\frac{L(xt)}{L(x)} \xrightarrow{x \rightarrow +\infty} 1$.

Then,

(1) For $a_n = \inf\{x : \mathbb{P}(|X_i| > x) \leq \frac{1}{n}\}$ and $b_n = n \mathbb{E}[X_i \mathbb{1}_{|X_i| \leq a_n}]$, we have

$$\frac{S_n - b_n}{a_n} \Rightarrow Y, \quad n \rightarrow +\infty$$

(2) The characteristic function of Y is $\Phi_Y(t) = \exp(i t c - b |t|^\alpha (1 + i k \operatorname{sgn}(t) \omega_\alpha(t)))$, where c and b are constants, $k = 2\theta - 1$, and

$$\omega_\alpha(t) = \begin{cases} \tan \frac{\pi\alpha}{2} & \alpha \neq 1 \\ \frac{2}{\pi} \log |t| & \alpha = 1 \end{cases}$$

We give two examples using this theorem.

5.2.1 Cauchy distribution

Define the Cauchy distribution as $f(x) = \frac{1}{\pi(1+x^2)}$, and the characteristic function is

$$\Phi_X(t) = \int_{-\infty}^{+\infty} \frac{e^{itx}}{\pi(1+x^2)} dx = \exp(-|t|)$$

Proposition 5.2.2.

If $X_1, X_2, \dots, X_n$ are i.i.d. symmetric random variables with distribution function continuous at 0, then

$$\frac{1}{n} \left(\frac{1}{X_1} + \dots + \frac{1}{X_n} \right) \Rightarrow \text{Cauchy distribution}$$

Proof. Take $\alpha = 1$, $k = 0$, $\theta = \frac{1}{2}$, and $Y_i = \frac{1}{X_i}$. Let f be the distribution function of X_i , then

$$\mathbb{P}(|Y_i| \geq x) = \int_0^{x^{-1}} f(s) ds = \frac{1}{x} \cdot \frac{\int_0^{x^{-1}} f(s) ds}{x^{-1}}$$

and since f is continuous at 0,

$$\frac{\int_0^{x^{-1}} f(s) ds}{x^{-1}} \rightarrow 1, \text{ as } x \rightarrow +\infty$$

Hence Y_i satisfy the conditions in the theorem above. As a result, as $n \rightarrow +\infty$, $a_n \Rightarrow n$, $b_n = 0$ and

$$\frac{1}{n} \left(\sum_{i=1}^n \frac{1}{X_i} \right) \Rightarrow Y$$

□

Corollary 5.2.3.

If X_i are independent random variables with Cauchy distribution, then $\frac{X_1 + \dots + X_n}{n}$ also has Cauchy distribution.

5.2.2 Asymmetric case

In the conditions of the theorem above, take $L = 1$ and ignore the limit in (a), i.e. for $x \geq 1$,

$$\mathbb{P}(X > x) = \theta x^{-\alpha}, \quad \mathbb{P}(X < -x) = (1 - \theta) x^{-\alpha}$$

Hence

$$a_n = n^{\frac{1}{\alpha}}$$

,

$$b_n = n \int_1^{n^{\frac{1}{\alpha}}} \frac{\alpha}{x^{\alpha+1}} x(2\theta - 1) dx \sim \begin{cases} cn & \alpha \in (1, 2) \\ cn \log n & \alpha = 1 \\ cn^{\frac{1}{\alpha}} & \alpha \in (0, 1) \end{cases}$$

5.3 Proof of Theorem 5.0.1

Proof. For c small enough, take $A = n^{\frac{1}{\alpha}} \lambda$, and then

$$\begin{aligned} \mathbb{P}\left(\frac{M_n}{S_n} > c\right) &\geq \mathbb{P}(M_n > cA, S_n < A) \\ &\geq 1 - \mathbb{P}(M_n < cA) - \mathbb{P}(S_n > A) \\ &= 1 - \mathbb{P}(M_n/n^{\frac{1}{\alpha}} > \lambda c) - \mathbb{P}\left(\frac{S_n - b_n}{n^{\frac{1}{\alpha}}} > \lambda - \delta\right) > c \end{aligned}$$

In the last inequality, use Frechet distribution to control $\mathbb{P}(M_n/n^{\frac{1}{\alpha}} > \lambda c)$, and use the theorem above to control $\mathbb{P}(\frac{S_n - b_n}{n^{\frac{1}{\alpha}}} > \lambda - \delta)$. □

5.4 Stable law

Definition 5.4.1 (Stable law).

Y is a **stable law** if and only if $\forall k \in \mathbb{N}$, $\exists a_k, b_k$ s.t. i.i.d. $Y_i \sim Y$ satisfy

$$\frac{\sum_{i=1}^k Y_i - b_k}{a_k} \stackrel{d}{\sim} Y$$

Remark 5.4.2.

If Y_i is already stable, then so is Y . An example of stable law is the Gaussian distribution.

Theorem 5.4.3.

Y is the limit in law of $\frac{\sum_{i=1}^k X_i - b_k}{a_k}$ for some i.i.d. random variables X_i if and only if Y is stable.

Example 5.4.4.

For random walk X_k , let $\tau_0 = 0$, and $\tau_i = \inf\{k : X_k = i\}$, $\delta \tau_i = \tau_i - \tau_{i-1}$, then

$$\frac{\sum_{i=1}^n \delta \tau_i}{n^2} \Rightarrow Y$$

Lecture 6

Power of Two Choices

Consider putting n balls into n boxes. Let $X_i = \#\{\text{balls in the box indexed by } i\}$ and $L_n = \max_{1 \leq i \leq n} X_i$. We wish to reduce the number L_n to avoid local anomalies.

6.1 Warm-up

We first try putting the balls in the following way: For every ball, sample uniformly the boxes until an empty box is chosen, and put this ball into this empty box. In this case $L_n = 1$.

Let T_k be the waiting time of the k -th ball, and we can give some estimations about T_k .

Proposition 6.1.1.

$$\mathbb{E} \left[\sum_{k=1}^n T_k \right] = n \left(\sum_{k=1}^n \frac{1}{k} \right) \approx n \log n$$

Proof. Observe that $T_k \sim \text{Geo} \left(1 - \frac{k-1}{n} \right)$ and that T_k are independent with $\mathbb{E}[T_k] = \frac{n}{n+1-k}$ $\square$

Proposition 6.1.2.

$$\mathbb{P} \left(\left| \sum_{k=1}^n T_k - n \log n \right| > c \log n \right) \approx n^{-c}$$

Proof. We will not prove this result but instead give a loser bound.

$$\text{Var} \left[\sum_{k=1}^n T_k \right] = \sum_{k=1}^n \text{Var} [T_k] = \sum_{k=1}^n \frac{(k-1)/n}{(1 - (k-1)/n)^2} < \sum_{k=1}^n \frac{n^2}{k^2} \sim O(n^2)$$

Hence by Chebyshev's inequality, take $c = \epsilon n$, and we have

$$\mathbb{P} \left(\left| \sum_{k=1}^n T_k - n \log n \right| > c \log n \right) \leq \frac{O(n^2)}{(c \log n)^2} \xrightarrow{n \rightarrow +\infty} 0$$

Hence

$$\mathbb{P} \left(\sum_{k=1}^n T_k \in ((1 - \epsilon) n \log n, (1 + \epsilon) n \log n) \xrightarrow{n \rightarrow +\infty} 0 \right)$$

□

6.2 Uniform allocation

Definition 6.2.1 (Uniform allocation).

A **uniform allocation** is performed by sampling independently for every ball a uniform index $i \in [n]$ and put the ball in the box indexed by i .

Theorem 6.2.2.

$$\mathbb{P} \left(L_n \asymp \frac{\log(n)}{\log(\log(n))} \right) \xrightarrow{n \rightarrow +\infty} 1$$

Proof. $X_i \sim \text{Bin}(n, \frac{1}{n})$, so $\mathbb{E}[X_i] = 1$, $\text{Var}[X_i] = n \cdot \frac{1}{n}(1 - \frac{1}{n}) = 1 - \frac{1}{n}$.

Step 1 Upper bound

First note that

$$\begin{aligned} \mathbb{P}(L_n \geq c) &= \mathbb{P}(\exists 1 \leq i \leq n, X_i \geq c) \\ &\leq \sum_{i=1}^n \mathbb{P}(X_i \geq c) = n \mathbb{P}(X_1 \geq c) \end{aligned}$$

Note that for large n , $\text{Bin}(n, \frac{1}{n}) \approx \text{Poisson}(1)$, which is the sum of n independent random variables with $\text{Poisson}(\frac{1}{n})$ distribution. The intuition is that for $Z \sim \text{Poisson}(\frac{1}{n})$, $\mathbb{P}(Z \geq 2) = 1 - e^{-1/n} - \frac{1}{n}e^{-1/n}$ is very small, so it is close to the Bernoulli($\frac{1}{n}$) distribution. Observe that $B(n, \frac{1}{n}) = \sum_{i=1}^n Y_i$ where $Y_i \sim \text{Bernoulli}(\frac{1}{n})$.

We now give a rigorous tail bound.

$$\begin{aligned} \mathbb{P}(X_i \geq c) &= \mathbb{P}(\exp(\lambda X_i) \geq \exp(\lambda c)) \\ &\leq e^{-\lambda c} \mathbb{E}[\exp(\lambda X)] = e^{-\lambda c} (\mathbb{E}[\exp(\lambda Y_i)])^n \\ &= e^{-\lambda c} (1 - 1/n + e^\lambda/n)^n \\ &\xrightarrow{n \rightarrow +\infty} e^{-\lambda c} \exp(e^\lambda - 1) \end{aligned}$$

Take $e^\lambda = c$, then $\mathbb{P}(L_n \geq c) \leq \exp(-c \log c + c - 1 + \log n)$. We want the RHS to tend to 0 as n tends to infinity.

Take θ slightly larger than 1, and let

$$c = \frac{\theta \log n}{\log(\log n)}$$

then

$$c \log c = \frac{\theta \log n}{\log(\log n)} (\log(\log n) - \log(\log(\log n)))$$

$$\mathbb{P}(L_n \geq c) \xrightarrow{n \rightarrow +\infty} 0$$

Step 2 Lower bound

We propose two methods of attaining a lower bound, indicating the first and giving a detailed explanation of the second.

The first attempt is to use the inclusion-exclusion principal:

$$\mathbb{P}(\cup_{i=1}^n \{X_i \geq c\}) \geq n \mathbb{P}(X_1 \geq c) - \frac{n(n-1)}{2} \mathbb{P}(\{X_1 \geq c\} \cap \{X_2 \geq c\})$$

Hope that $\mathbb{P}(X_i > c) \sim \frac{1}{n}$, then RHS is approximately larger than $\frac{1}{2}$.

We now concentrate on the second line of thought, in which we seek aid from independence. Due to superposition properties of Poisson processes, we have the following lemma:

Lemma 6.2.3.

Apply uniform allocation to $\text{Poisson}(M)$ balls, then $X_i \sim \text{Poisson}(\frac{M}{n})$ are i.i.d. random variables.

Remark 6.2.4.

In this case, L_n has the Gumbel distribution.

Special case with Poisson points

Run uniform allocation to $\text{Poisson}(M)$ and let $\lambda = \frac{M}{n}$. Then by Lemma, the corresponding $(\tilde{X}_i)_{1 \leq i \leq n} \sim \text{Poi}(\frac{M}{n})$ are i.i.d. Let $\tilde{L}_n = \max_i \tilde{X}_i$, then

$$\begin{aligned} \mathbb{P}(\tilde{L}_n > c) &= 1 - \mathbb{P}(\tilde{L}_n \leq c) \\ &= 1 - \mathbb{P}(\cap_{i=1}^n \{\tilde{X}_i \leq c\}) \\ &\stackrel{\text{independence}}{=} 1 - \mathbb{P}(\tilde{X}_1 \leq c)^n \\ &= 1 - (1 - \mathbb{P}(\tilde{X}_1 > c))^n \approx 1 - (1 - e^{-\lambda} \frac{\lambda^c}{c!})^n \\ &\approx 1 - \exp(-n e^{-\lambda} \frac{\lambda^c}{c!}) \\ &\stackrel{\text{take } \lambda=1/2}{=} 1 - \exp(-e^{\log n - 1/2 + c \log \frac{1}{2} - c \log c}) \end{aligned}$$

where we used the approximation $\mathbb{P}(\tilde{X}_i > c) = \sum_{k=c}^{+\infty} e^{-\lambda} \frac{\lambda^k}{k!} \approx e^{-c} \frac{\lambda^c}{c!}$.

Take $c = \theta \frac{\log(n)}{\log(\log(n))}$, where θ is slightly smaller than 1, then

$$\mathbb{P}(\tilde{L}_n > c) \xrightarrow{n \rightarrow +\infty} 1$$

Genral case with amount of points n

Put a natural monotone coupling between $(\tilde{X}_i)$, (X_i) , and let $N \sim \text{Poisson}(n/2)$, then

$$\begin{aligned}
\mathbb{P}(L_n \geq \theta \log n / \log(\log n)) &= \mathbb{P}(L_n \geq \frac{\theta \log n}{\log(\log n)}, N < n) + \mathbb{P}(L_n \geq \frac{\theta \log n}{\log(\log n)}, N \geq n) \\
&\geq \mathbb{P}(\tilde{L}_n \geq \frac{\theta \log n}{\log(\log n)}, N < n) \quad \text{since } L_n \geq \tilde{L}_n \\
&= \mathbb{P}(\tilde{L}_n \geq \frac{\theta \log n}{\log(\log n)}) - \mathbb{P}(\tilde{L}_n \geq \frac{\theta \log n}{\log(\log n)}, N \geq n) \\
&\geq \mathbb{P}(\tilde{L}_n \geq \frac{\theta \log n}{\log(\log n)}) - \mathbb{P}(N \geq n) \\
&\xrightarrow{n \rightarrow +\infty} 1
\end{aligned}$$

where the last line is due to the fact that the first term tends to 1 and the second term tends to 0 by Chebyshev:

$$\mathbb{P}(N \geq n) \leq \frac{\text{Var}[N]}{(n/2)^2} = \frac{2}{n} \xrightarrow{n \rightarrow +\infty} 0$$

In summart, the idea of the coupling is that we first use the uniform allocation to Poisson $(n/2)$ balls, and if the number of balls determined by the Poisson distribution is no more than n , we deal with the remaining balls using the original uniform allocation. $\square$

6.3 Power of two choices

Definition 6.3.1 (Greedy (d) algorithm).

Choose uniformly d boxes independently and put the ball in the box wit the smallest number of balls.

Theorem 6.3.2.

Using the Greedy (d) algorithm defined above, with high probability

$$L_n \simeq \frac{\log(\log n)}{\log d}$$

Remark 6.3.3.

The uniform allocation is actually taking $d = 1$ in this case, and we get $L_n \simeq \log n$. The intuition of this result is that the more choices the better.

Proof. **Heuristics**

Let β_i be the "typical upper bound" of boxes with more than i balls, then

$$\beta_{i+1} \leq n \left(\frac{\beta_i}{n} \right)^d$$

Intuitively, consider the first time there are $i + 1$ balls, there must be some time when the i balls in this box become $i + 1$ balls. This can happen at the time the k -th ball is put into a box, $k = 1, \dots, n$, which gives n choices. If so, all the d boxes chosen must have $\geq i$ balls, which contributes to $(\frac{\beta_i}{n})^d$.

From this inequality, we get

$$\beta_0 = n, \beta_1 \leq n, \beta_2 \leq \frac{n}{2}, \quad \frac{\beta_{i+1}}{n} \leq \left(\frac{\beta_i}{n} \right)^d, \log\left(\frac{\beta_{i+1}}{n}\right) \leq d \log\left(\frac{\beta_i}{n}\right)$$

$$\beta_k \leq n \left(\frac{1}{2} \right)^{d^{k-2}}$$

$$n \left(\frac{1}{2} \right)^{d^{k-2}} = 1 \Rightarrow k \simeq \frac{\log(\log n)}{\log d}$$

Rigorous proof

We now give a rigorous proof. Let $B_i(t)$ be the number of boxes with at least i balls at time t , $0 \leq t \leq n$. Let $B_i = B_i(n)$, and let $\mathcal{F}_i(t) = \sigma(B_j(t), 1 \leq j \leq i)$. The key observation is that

$$\mathbb{E}[B_{i+1}(t+1) | \mathcal{F}_i(t)] \leq B_{i+1}(t) + \left(\frac{B_i(t)}{n} \right)^d$$

From this we derive that

$$\mathbb{E}[B_{i+1}(n)] \leq \sum_{t=0}^{n-1} \mathbb{E}\left[\left(\frac{B_i(t)}{n}\right)^d\right] \leq n \mathbb{E}\left[\left(\frac{B_i}{n}\right)^d\right]$$

Object : show that $\beta_{i+1} \leq e \cdot n \cdot \left(\frac{\beta_i}{n}\right)^d$ is an upper bound of B_i .

Let $E_i = \{B_i(n) \leq \beta_i\}$.

Step 1 : $\forall \beta_i \geq n \left(\frac{4 \log n}{n} \right)^{1/d}$, we have

$$\mathbb{P}(E_{i+1}^c | E_i) \leq \frac{1}{n^2} \frac{1}{\mathbb{P}(E_i)}$$

Step 1+ : Under the assumption of Step 1, $\mathbb{P}(E_i^c) \leq \frac{i+1}{n^2}$ since

$$\begin{aligned} \mathbb{P}(E_{i+1}^c) &= \mathbb{P}(E_i \cap E_{i+1}^c) + \mathbb{P}(E_i^c \cap E_{i+1}^c) \\ &\leq \mathbb{P}(E_{i+1}^c | E_i) \mathbb{P}(E_i) + \mathbb{P}(E_i^c) \\ &\stackrel{\text{induction hypothesis}}{\leq} \frac{1}{n^2} + \frac{i+1}{n^2} = \frac{i+2}{n^2} \end{aligned}$$

Now we return to proving step 1 : Let $Y_j \sim \text{Bernoulli}((\frac{\beta_i}{n})^d)$ be i.i.d random variables, then

$$\begin{aligned} \mathbb{P}(E_{i+1}^c | E_i) &= \frac{1}{\mathbb{P}(E_i)} \mathbb{P}(E_{i+1}^c \cap E_i) \\ &\leq \frac{1}{\mathbb{P}(E_i)} \mathbb{P}\left(\sum_{j=1}^n Y_j > \beta_{i+1}\right) \\ &\stackrel{\text{Hoeffding}}{\leq} \frac{1}{\mathbb{P}(E_i)} \exp\left(-n \left(\frac{\beta_i}{n}\right)^d / 2\right) \\ &\leq \frac{1}{\mathbb{P}(E_i)} \exp(-2 \log n) = \frac{1}{n^2} \frac{1}{\mathbb{P}(E_i)} \end{aligned}$$

Step 2 : Let $k_0 = \inf_k \{ \beta_k \leq n (\frac{4 \log n}{n})^{1/d} \}$. The method is similar to that of Step 1 and is left for the student to consider.

Here $\beta_{k_0+1} = 100 \log n$, $\beta_{k_0+2} \leq (\frac{1-\log n}{n})^{\frac{1}{d}} \leq \frac{1}{2}$.

Denote $X = \text{Bin}(n, (\frac{\beta_{k_0+1}}{n})^d)$ and $p_n = \frac{\log n}{n}$, then

$$\begin{aligned} &\mathbb{P}\left(\text{Bin}\left(n, \left(\frac{\beta_{k_0+1}}{n}\right)^d\right) > \beta_{k_0+2}\right) \\ &\leq \mathbb{P}(\exp(\lambda X) > \exp(\lambda \beta_{k_0+2})) \\ &\leq \exp(-\lambda \beta_{k_0+2}) \mathbb{E}[e^{\lambda X}] \\ &= \exp(-\lambda \beta_{k_0+2}) (e^\lambda p_n + 1 - p_n)^n \\ &\leq \exp\left(-\frac{1}{2} \lambda\right) \exp(n p_n (e^\lambda - 1)) \\ &\leq \exp\left(-\frac{1}{2} \lambda + n p_n (\lambda + \lambda^2/2)\right) \end{aligned}$$

Minimizing the last line yields that when taking $\lambda = \frac{1}{2np_n}$,

$$\mathbb{P}\left(\text{Bin}\left(n, \left(\frac{\beta_{k_0+1}}{n}\right)^d\right) > 1/2\right) \leq \exp\left(-\frac{1/4}{2np_n}\right) = \exp\left(-\frac{1}{8np_n}\right)$$

□

Lecture 7

Random graphs: Erdos-Renji graph; preferential attachment

7.1 Preferential attachment

Construct a random graph with preferential attachment in the following way:

1. Add vertice i to the graph.
2. Pick a unique $j < i$ to connect with i , each j being picked with probability

$$p_j = \frac{\deg(j)}{\sum_{k=1}^{i-1} \deg(k)}$$

3. Back to Step 1 and start dealing with vertice $i + 1$.

Theorem 7.1.1.

Define $p_k^{(n)} = \frac{1}{n} \#\{1 \leq i \leq n : \deg(i) = k\}$, then,

$$p_k^{(n)} \xrightarrow{n \rightarrow +\infty} \propto k^{-3}$$

Proof. Denote $D_t(i)$ be degree of i at time t , and do continuation by letting

$$\partial_t D_t(i) = \frac{D_t(i)}{2t}$$

Then

$$\partial_t \log D_t(i) = \frac{1}{2t}$$
$$D_t(i) = \left(\frac{t}{t_i}\right)^{\frac{1}{2}}$$

where t_i is birth time of i .

Hence,

$$\#\{i : D_t(i) \in [k, k+1)\} \sim \frac{\frac{t}{k^2} - \frac{t}{(k+1)^2}}{t} \propto k^{-3}$$

□

Example 7.1.2 (Unifrom attachment).

Usually when there is not much additional information about the vertices, we would just let $p_j = \frac{1}{i-1}$.

7.2 Erdos-Renji graph

Definition 7.2.1.

An **Erdos-Renji graph** is a graph $G(n, p)$ with n vertices, and every edge between two vertices i and j has a probability p to be connected.

The question arises that how big should p to ensure that the graph is connected? The theorem below gives an estimation.

Theorem 7.2.2.

$$\begin{aligned} \mathbb{P}(G(n, \frac{1+\epsilon}{n}) \text{ connected}) &\xrightarrow{n \rightarrow +\infty} 1 \\ \mathbb{P}(G(n, \frac{1-\epsilon}{n}) \text{ connected}) &\xrightarrow{n \rightarrow +\infty} 0 \end{aligned}$$

Proof. To prove the graph is not connected, it suffices to find an isolated point. We claim that

$$\mathbb{P}\{\text{isolated points} > 1\} = \begin{cases} \rightarrow 1 & p = \frac{(1-\epsilon) \log(n)}{n} \\ \rightarrow 0 & p = \frac{(1+\epsilon) \log(n)}{n} \end{cases}$$

To see this observe that

$$\mathbb{P}\{\text{isolated points} > 1\} \tag{7.1}$$

$$\leq \mathbb{E}[\# \text{ isolated vertices}] = \mathbb{E}\left[\sum_{v \in G} 1_{\{v \text{ is an isolated vertex}\}}\right] \tag{7.2}$$

$$= \sum_{v \in G} \mathbb{P}(v \text{ isolated}) = n(1-p)^{n-1} = \exp(\log(n) - n \log(n)/n(1+\epsilon)) \rightarrow 0. \tag{7.3}$$

for $p = \frac{(1+\epsilon) \log(n)}{n}$ In the case that $p = \frac{(1-\epsilon) \log(n)}{n}$, we have, using the same derivation that

$$\mathbb{E}[\# \text{ isolated vertices}] = n^\epsilon$$

Below we list some inequalities we can use:

* Chebyshev: $\mathbb{P}(|X - \mathbb{E}[X]| \geq \delta \mathbb{E}[X]) \leq \frac{\text{Var}(X)}{\delta^2 \mathbb{E}[X]^2}$

* Paley-Zygmund $\mathbb{P}(X \geq \lambda \mathbb{E}[X]) \geq (1 - \lambda)^2 \mathbb{E}[X]^2 / \mathbb{E}[X^2]$ which is a one-sided lower bound.

In this case, writing X for the number of isolated vertices,

$$\mathbb{E}[X^2] = \mathbb{E}[X] + \mathbb{E}[X(X-1)] \quad (7.4)$$

$$= \mathbb{E}[X] + \sum_{i \neq j} \mathbb{E}[1_{\{i \text{ is isolated}\}} 1_{\{j \text{ is isolated}\}}] \quad (7.5)$$

$$= n^\epsilon + n(n-1)(1-p)^{2n-3} \approx n^\epsilon + n^{2\epsilon} \quad (7.6)$$

thus $\text{Var}(X) \approx n^\epsilon$. Thus by Chebyshev's inequality, we have

$$\mathbb{P}(X = 0) \leq \frac{\text{Var}(X)}{\mathbb{E}[X]^2} \approx n^{-\epsilon} \rightarrow 0 \quad (n \rightarrow +\infty)$$

. We now use this to prove that in $G(n, p)$, if $p_n = (1 + \epsilon) \frac{\log(n)}{n}$, then

$$\mathbb{P}(G(n, p_n) \text{ is connected}) \rightarrow 1 \quad (n \rightarrow +\infty)$$

Note that this implies that the threshold of connectivity is equal to the threshold of isolated points. We showed above that whp there are no isolated points. If the graph is still not connected, it can be partitioned into disjoint clusters, the smaller of which has size smaller than $n/2$. We consider the case where the vertices can be partitioned into two disjoint sets A and B , where $2 \leq \#A \leq \#B$, A is connected and there are no edges between A and B (B may have multiple clusters).

$$\mathbb{P}(\exists \text{ small clusters}) \quad (7.7)$$

$$= \sum_{k=2}^{\lfloor n/2 \rfloor} \mathbb{P}(1_{\{\#A=k\}}) \quad (7.8)$$

$$\leq \sum_{k=2}^{\lfloor n/2 \rfloor} \binom{n}{k} \mathbb{P}(\{1, \dots, k\} \text{ is connected}) \mathbb{P}(\text{no edges between } \{1, \dots, k\} \text{ and } \{k+1, \dots, n\}) \quad (7.9)$$

$$= \sum_{k=2}^{\lfloor n/2 \rfloor} \binom{n}{k} (1-p)^{k(n-k)} \mathbb{P}(\{1, \dots, k\} \text{ connected}) \quad (7.10)$$

$$\leq \sum_{k=2}^{\lfloor n/2 \rfloor} \binom{n}{k} (1-p)^{k(n-k)} k^{k-2} p^{k-1} \xrightarrow{n \rightarrow +\infty} 0 \quad \text{when } \frac{2 \log(n)}{3n} < p < \frac{2 \log(n)}{n}. \quad (7.11)$$

where in deriving the last inequality, we used the fact that

$$\mathbb{P}(\{1, \dots, k\} \text{ connected}) = \mathbb{P}(\exists \text{ spanning tree}) \leq \# \text{spanning trees} \cdot p^{k-1} = k^{k-2} p^{k-1}$$

□

We now make a digression and talk about a problem that appeared in the midterm.

Proposition 7.2.3.

$p = \frac{\log(n)+c}{n}, c > 0$ then $\mathbb{P}(X = 0) \xrightarrow{n \rightarrow +\infty} e^{-e^{-c}}$. In fact we have $\mathbb{P}(X = k) \rightarrow e^{-\lambda} \lambda^k / k!, \lambda = e^{-c}$.

To prove this, we use

Proposition 7.2.4 (moment criterion).

(X_n) positive random variables $\mathbb{E}[X_n^k] \xrightarrow{n \rightarrow +\infty} C_k \leq M^k k!$, then $X_n \xrightarrow{d} X$ with $\mathbb{E}[X^k] = C_k$.

Proof. $\mathbb{E}[X_n] \rightarrow C_1$ implies tightness, thus the sequence X_n admits weak convergence to $\tilde{X}$ along subsequences. $\mathbb{E}[e^{\lambda \tilde{X}}] = \sum_{k=0}^{+\infty} \mathbb{E}[\frac{(\lambda \tilde{X})^k}{k!}]$. We claim that

$$\mathbb{E}[\tilde{X}^k] = \lim_{n \rightarrow +\infty} \mathbb{E}[X_n^k] = C_k$$

To prove the claim, one method is to use cutoff long with some higher moment bounds. Alternatively, use Skorohod embedding in a space $\tilde{P}$ so that we obtain $X_n \xrightarrow{a.s.} \tilde{X}$. The uniform $L^{k+\epsilon}$ -moment gives uniform integrability. Thus $\mathbb{E}[X_n^k] \rightarrow \mathbb{E}[\tilde{X}^k](n \rightarrow +\infty)$. □

Proof of convergence to Poisson. As we calculated in the midterm,

$$\mathbb{E}[X] \rightarrow e^{-c}, \mathbb{E}[X(X-1)] \rightarrow e^{-2c}, \dots, \mathbb{E}[X(X-1) \dots (X-k+1)] \rightarrow e^{-kc}$$

thus the k -th moments converge to those of Poisson. By the moment criterion the limit is Poisson. □

Apart from connectivity we can also consider giant clusters. Consider $G(n, \lambda/n)$ for λ constant. Observe that in this case there are a lot of isolated points.

Moreover, $\#\{i \in V, i \sim 1\} \xrightarrow{d} Poi(\lambda)$ so $G(n, \lambda/n)$ looks like a Galton-Watson tree with $Poi(\lambda)$ production. We rank the cluster sizes

$$C_1^{\max} \geq C_2^{\max} \geq \dots$$

Theorem 7.2.5. * $0 < \lambda < 1$, then $C_1^{\max} \simeq \log(n)$.

* $\lambda > 1$, then $C_1^{\max} \simeq n$, $C_2^{\max} \simeq \log(n)$.

* $\lambda = 1$, then $(n^{-2/3} C_i^{\max}(n, 1/n))_i$ converges in law to a vector in ℓ^2 .

Reference: "scaling limit of critical Erdos Renyi graph".

Lecture 8

Simulation of SDE

The final goal of the remaining lectures is to understand (simulate) the diffusion model.

8.1 Brownian motion

Definition 8.1.1.

Standard Brownian motion $(W_t)_{t \geq 0}$ is a continuous process with centered independent stationary increments such that

* $W_0 = 0$.

* $W_t - W_s \stackrel{d}{\sim} N(0, t - s) (t > s)$ and $(W_{t_1}, W_{t_2} - W_{t_1}, \dots, W_{t_n} - W_{t_{n-1}}) (0 < t_1 < t_2 < \dots, t_n)$ are independent.

* $\forall w, t \rightarrow W_t(w)$ is continuous.

We remark that Brownian motion exists.

Proposition 8.1.2 (Donsker's invariance principle).

$(X_i)_{i \in \mathbb{N}}$ i.i.d random variables, $\mathbb{P}(X_i = 1) = \mathbb{P}(X_i = -1) = 1/2$, $S_n = \sum_{i=1}^n X_i$. Then

$$(\frac{1}{\sqrt{n}} S_{[nt]})_{t \geq 0} \xrightarrow{d} (W_t)_{t \geq 0}$$

In fact this result holds for any sequence of iid random variables with mean 0 and variance 1. Moreover, this is the easiest way to simulate Brownian motion, also called forward simulation. Alternatively, one can use backward simulation, using the Brownian bridge (or Levy's construction of Brownian motion).

Lemma 8.1.3.

Conditioning on the values $W_{t_i}, W_{t_{i+1}}$, sample

$$W_{\frac{t_i+t_{i+1}}{2}} = 1/2(W_{t_i} + W_{t_{i+1}}) + N(0, \frac{1}{4}(t_{i+1} - t_i))$$

where the normal summand is independent of $W_{t_i}, W_{t_{i+1}}, \dots$, that is, the known information. Abbreviate $\tilde{W} = W_{\frac{t_i+t_{i+1}}{2}}, \Delta t_i = t_{i+1} - t_i, \Delta W_{t_i} = W_{t_{i+1}} - W_{t_i}$.

$$\mathbb{P}(W_{t_1} = x_1, W_{t_2} = x_2, \dots, W_{t_{i-1}} = x_{i-1}, \dots) = \mathbb{P}(W_{t_1} = x_1, \Delta W_{t_1} = x_2 - x_1, \dots, \Delta W_{t_i} = x_{i+1} - x_i, \dots) \quad (8.1)$$

$$= P_{t_1}(x_1)P_{\Delta t_1}(x_2 - x_1) \dots P_{\Delta t_i}(x_{i+1} - x_i) \dots \quad (8.2)$$

Thus

$$\mathbb{E}[f(W_{t_1}, W_{t_2}, \dots, W_{t_i}, \tilde{W}, W_{t_{i+1}}, \dots) | W_{t_1} = x_1, W_{t_2} = x_2, \dots, W_{t_i} = x_i, W_{t_{i+1}} = x_{i+1}, \dots] \quad (8.3)$$

$$= \frac{\int_{\mathbb{R}} f(x_1, x_2, \dots, x_i, y, x_{i+1}, \dots) p_{t_1}(x_1) p_{\Delta t_1}(x_2 - x_1) \dots p_{1/2\Delta t_i}(y - x_i) p_{1/2\Delta t_i}(x_{i+1} - y) \dots dy}{\mathbb{P}(W_{t_1} = x_1, W_{t_2} = x_2, \dots, W_{t_{i-1}} = x_{i-1}, W_{t_i} = x_i \dots)} \quad (8.4)$$

$$= \frac{1}{p_{\Delta t_i}(x_{i+1} - x_i)} \int_{\mathbb{R}} f(x_1, x_2, \dots, x_i, y, x_{i+1}, \dots) p_{1/2\Delta t_i}(y - x_i) p_{1/2\Delta t_i}(x_{i+1} - y) dy \quad (8.5)$$

and the density function is $\frac{1}{\sqrt{2\pi\Delta t_i/4}} \exp(-\frac{|y - \frac{(x_i+x_{i+1})}{2}|^2}{2\Delta t_i/4})$

8.2 Ito integral and Ito formula

Definition 8.2.1 (Ito integral for BM).

$$\int_0^t f(s, W_s) dW_s = \lim_{|\Delta h|=t/h \rightarrow 0} \sum_{i=0}^n f(ih, W_{ih})(W_{(i+1)h} - W_{ih})$$

where convergence is in the sense of L^2 .

Note that the integral also makes sense for semi-martingales.

Definition 8.2.2 (Quadratic variation).

$$< X >_t = \lim_{|\Delta h| \rightarrow 0} \sum_{i=0}^{n-1} (X_{(i+1)h} - X_{ih})^2$$

All semi-martingales have quadratic variation. For Brownian motion, we have $< W >_t = t$.

Theorem 8.2.3 (Ito's formula).

X_t is a semi-martingale and f is differentiable, then

$$df(t, X_t) = \partial_t f(t, X_t) dt + \partial_x f(t, X_t) dX_t + \frac{1}{2} \partial_{xx} f(t, X_t) d< X >_t$$

8.3 SDE simulation

Euler-Maruyama algorithm:

$$X_t^{(h)} = X_{ih}^{(h)} + b(ih, X_{ih}^{(h)})(t - ih) + \sigma(ih, X_{ih}^{(h)})(W_t - W_{ih}), X_0^{(h)} = X_0$$

Theorem 8.3.1.

Assume that (1) $|b(t, x) - b(s, y)| + |\sigma(t, x) - \sigma(s, y)| \leq C(|x - y| + |t - s|^{1/2})$ and (2) $|b(0, 0) + c(0, 0)| \leq c$, then

$$\mathbb{E}[\sup_{0 \leq t \leq T} |X_t^{(h)} - X_t|^p] \leq h^{p/2}$$

Lecture 9

Langevin dynamic

9.1 Fokker-Plank equation

Definition 9.1.1.

$$dX_t = -\nabla V(X_t) dt + \sqrt{2} dW_t \in \mathbb{R}^d$$

where W is Brownian Motion, and $V \in \mathbb{R}^d \rightarrow \mathbb{R}$ is potential function.

Object

As t tends to infinity, $X_t \implies X_\infty \sim \frac{1}{Z} e^{-V}$ which is the Boltzmann distribution.

Proposition 9.1.2.

Let q_t be the distribution of X_t , then

$$\partial_t q_t = \Delta q_t + \nabla \cdot (q_t \nabla V)$$

Proof. $\forall$ test function f , we have

$$\mathbb{E}[f(X_t)] = \int_{\mathbb{R}^d} f q_t$$

$$\mathbb{E}[\partial_t f(X_t)] = \int_{\mathbb{R}^d} f \partial_t q_t$$

By Itô's formula,

$$df(X_t) = \nabla f(X_t) \cdot dX_t + \frac{1}{2} \Delta f(X_t) d\langle X \rangle_t$$

By $\langle X \rangle_t = 2\langle W \rangle_t = 2t$,

$$\begin{aligned} df(X_t) &= \nabla f(X_t) \cdot \left(-\nabla V(X_t) dt + \sqrt{2} dW_t \right) + \Delta f(X_t) dt \\ &= \left(-\nabla f(X_t) \cdot \nabla V(X_t) + \Delta f(X_t) \right) dt + \sqrt{2} \nabla f(X_t) dW_t \end{aligned}$$

So

$$\begin{aligned}
\mathbb{E}[\partial_t f(X_t)] &= \int_{\mathbb{R}^d} (-\nabla f \cdot \nabla V + \Delta f) q_t \\
&= \int_{\mathbb{R}^d} \Delta q_t f + \int_{\mathbb{R}^d} (-\nabla f) \cdot (q_t \nabla V) \quad (\text{By Itô's integral this term under } \mathbb{E}[\cdot] = 0) \\
&= \int_{\mathbb{R}^d} \left(\Delta q_t + \nabla \cdot (q_t \nabla V) \right) f
\end{aligned}$$

Hence

$$\partial_t q_t = \Delta q_t + \nabla \cdot (q_t \nabla V)$$

□

Proposition 9.1.3.

$\pi = \frac{1}{Z} e^{-V}$ is the stationary measure.

Proof. It suffices to justify that $\mathbb{E}_\pi[f(X_t)] = \mathbb{E}_\pi[f(X_0)]$ for any test function f .

Hence, it suffices to justify that

$$\Delta q_t + \nabla \cdot (q_t \nabla V) = 0 \quad \text{if} \quad q_0 \sim \pi$$

or equivalently that

$$\partial_t q_t = \Delta q_t + \nabla \cdot (q_t \nabla V) = 0 \quad \text{if} \quad q_t \equiv \pi$$

$$\text{Indeed. } \Delta(e^{-V}) + \nabla \cdot (e^{-V} \nabla V) = -\nabla \cdot (e^{-V} \nabla V) + \nabla \cdot (e^{-V} \nabla V) = 0$$

□

9.2 Poincare inequality

Now we come up naturally with the question about the rate of convergence $\rho_t \rightarrow \pi$. In order that the rate be not too slow, we need to add some conditions to V .

Definition 9.2.1 (Poincaré inequality).

$$\text{Var}_\pi(f) \leq \frac{1}{\lambda} \mathcal{E}(f, f), \quad \mathcal{E}(f, f) = \mathbb{E}_\pi[f \cdot (-\mathcal{L})f]$$

Lemma 9.2.2.

In Langevin dynamics, $\mathbb{E}_\pi[f(-\mathcal{L}f)] = \mathbb{E}_\pi[|\nabla f|^2]$

Proof.

$$df(X_t) \stackrel{\text{Itô}}{=} \nabla f(X_t) dX_t + \frac{1}{2} \Delta f(X_t) d\langle X \rangle_t$$

$$\begin{aligned}
&= -\nabla f(X_t) \cdot \nabla V(X_t) dt + 2 \cdot \frac{1}{2} \Delta f(X_t) dt + \dots dW_t \\
&\implies \mathcal{L}f = -\nabla f \cdot \nabla V + \Delta f
\end{aligned}$$

Integration by parts

$$\begin{aligned}
\mathbb{E}_\pi [f(-\mathcal{L}f)] &= \int_{\mathbb{R}^d} f(\nabla f \cdot \nabla V - \Delta f) \frac{1}{Z} e^{-V} \\
&= \int_{\mathbb{R}^d} f \nabla f \cdot \nabla V \frac{1}{Z} e^{-V} + \nabla f \cdot \nabla \left(\frac{1}{Z} e^{-V} f \right) \\
&= \int_{\mathbb{R}^d} |\nabla f|^2 \frac{1}{Z} e^{-V} + f \nabla f \cdot (-\nabla V) \frac{1}{Z} e^{-V} \\
&= \int_{\mathbb{R}^d} |\nabla f|^2 \frac{1}{Z} e^{-V} = \mathbb{E}_\pi [|\nabla f|^2]
\end{aligned}$$

□

Remark 9.2.3.

A **tensorization** is a trick that is often used:

For independent r.v. X_1 and X_2 , we have

$$\begin{aligned}
\text{Var}[f(X_1)] &\leq \lambda_1 \mathbb{E} [|\nabla f(X_1)|^2] \\
\text{Var}[f(X_2)] &\leq \lambda_2 \mathbb{E} [|\nabla f(X_2)|^2]
\end{aligned}$$

Hence

$$\Rightarrow X_1 \perp X_2, \quad X_1 + X_2 \quad \text{s.t.} \quad \text{Var}[f(X_1 + X_2)] \leq (\lambda_1 + \lambda_2) \mathbb{E} [|\nabla f(X_1 + X_2)|^2]$$

Proof. Via Efron-Stein inequality, we get that for $X'_1 \perp X'_2$ s.t. $X'_1 \stackrel{d}{\sim} X_1$, $X'_2 \stackrel{d}{\sim} X_2$ and $X'_1 \perp X_1$, $X'_2 \perp X_2$

$$\begin{aligned}
\text{Var}[f(X_1, X_2)] &\leq \mathbb{E} [(f(x_1, x_2) - f(x_1, x'_2))^2] + \mathbb{E} [(f(x_1, x_2) - f(x'_1, x_2))^2] \\
&\leq \text{Var}_{X_1}(f | X_2) + \text{Var}_{X_2}(f | X_1)
\end{aligned}$$

Here let $f(X_1 + X_2) = f(X_1, X_2)$.

□

Lemma 9.2.4.

If Poincare inequality holds, then

$$\text{Var}_\pi [P_t f] \leq e^{-2\lambda t} \text{Var}_\pi [f]$$

where $P_t = e^{Lt}$, $P_t f(x) = \mathbb{E} [f(X_t) | X_0 = x]$, and L is defined as $LP_t f(x) = \frac{d\mathbb{E}[f(X_t)|X_0=x]}{dt}$.

Proof. First by invariance of measure π ,

$$\mathbb{E}_\pi [f(X_t)] = \mathbb{E}_\pi [P_t f] = \mathbb{E}_\pi [L f] \triangleq m$$

Then

$$\begin{aligned} \frac{d}{dt} \text{Var}_\pi(P_t f) &= \frac{d}{dt} \mathbb{E}_\pi [(P_t f - m)^2] \\ &= \frac{d}{dt} \int \mathbb{E} [(P_t f - m)^2 \mid X_0 = x] \pi(x) dx \\ &= 2 \mathbb{E}_\pi [(P_t f - m)(L P_t f)] \\ &= 2 \mathbb{E}_\pi [(P_t f)(L P_t f)] - m \frac{d}{dt} \mathbb{E}_\pi [P_t f] \\ &= -2 \mathcal{E}(P_t f, P_t f) \end{aligned}$$

the last equation due to $\mathbb{E}_\pi [P_t f] = \mathbb{E}_\pi [f]$, which is constant for all t .

By Poincare inequality, we have

$$\mathcal{E}(P_t f, P_t f) \geq \lambda \text{Var}_\pi(P_t f)$$

Hence

$$\frac{d}{dt} \text{Var}_\pi(P_t f) \leq -2\lambda \text{Var}_\pi(P_t f) \implies \text{Var}_\pi(P_t f) \leq e^{-2\lambda t} \text{Var}_\pi(f)$$

□

Theorem 9.2.5.

Suppose $X_0 \stackrel{d}{\sim} \rho_0 \in M_1(\mathbb{R}^d)$, with Radon-Nikodym derivative $h_t = \frac{d\mu_0}{d\pi}$. Let $X_t \stackrel{d}{\sim} \rho_t$, then:

$$\mathbb{E}_\pi [|h_t - 1|^2] \leq e^{-2\lambda t} \mathbb{E}_\pi [|h_0 - 1|^2]$$

Proof. Claim that $h_t = P_t h_0$.

Indeed.

$$\mathbb{E} [f(X_t)] = \int f \rho_t dx = \int f \frac{d\rho_t}{d\pi} d\pi = \int f h_t d\pi = \mathbb{E}_\pi [f h_t] \quad \text{--- } \textcircled{A}$$

While on the other hand, since p_t is reversible with π , we have

$$\begin{aligned} \mathbb{E} [f(X_t)] &= \int [\mathbb{E} [f(X_t) \mid X_0 = x]] d\rho_0(X) = \int (P_t f) h_0 d\pi = \mathbb{E}_\pi [h_0 P_t f] \\ &= \mathbb{E}_\pi [f(P_t h_0)] \quad \text{by } P_t \quad \text{--- } \textcircled{A}' \end{aligned}$$

Hence $\textcircled{A} = \textcircled{A}'$ gives $\mathbb{E}_\pi [h_t] = 1$.

Moreover, since $\mathbb{E}_\pi [h_t] = 1$, we get using the previous lemma:

$$\mathbb{E}_\pi [|h_t - 1|^2] = \text{Var}_\pi(h_t) = \text{Var}_\pi(P_t h_0) \leq e^{-2\lambda t} \text{Var}_\pi(h_0)$$

□

9.3 Log-Sobolev inequality

Definition 9.3.1 (Entropy and relative entropy).

For measures $\mathbb{P} \sim p$ and $\mathbb{Q} \sim q$ on space $\mathcal{X}$, the **Entropy** and **Relative Entropy** are:

$$H(\mathbb{P}) = \sum_x p(x) \log p(x) \quad \text{or} \quad \int_{\mathcal{X}} p(x) \log p(x) dx$$

$$H(\mathbb{P} \parallel \mathbb{Q}) = KL(\mathbb{P} \parallel \mathbb{Q}) = \int_{\mathcal{X}} \frac{d\mathbb{P}}{d\mathbb{Q}} \log \left(\frac{d\mathbb{P}}{d\mathbb{Q}} \right) d\mathbb{Q}$$

Proposition 9.3.2.

For entropy and relative entropy, we have the following properties:

1. $H(\mathcal{P}), H(\mathcal{P} \parallel \mathcal{Q}) \geq 0$
2. $\mathcal{P} = \mathcal{Q} \implies H(\mathcal{P} \parallel \mathcal{Q}) = 0$

Proof. Denote the convex function $h(x) = x \log x$, and define $f = \frac{d\mathbb{P}}{d\mathbb{Q}}$.

Then by Jensen's inequality

$$H(\mathbb{P} \parallel \mathbb{Q}) = \int h(f(x)) d\mathbb{Q} \geq h \left(\int f d\mathbb{Q} \right) = h(1) = 0$$

$$H(\mathbb{P}) = \int h(p) dx \geq h \left(\int p dx \right) = h(1) = 0$$

□

Theorem 9.3.3 (Log-Sobolev inequality).

$$H(X_t \parallel \pi) \leq e^{-16\lambda t} H(X_0 \parallel \pi) \quad (X_t \sim \rho_t)$$

$$\|p_t - \pi\|_{TV} \leq C e^{-8\lambda t}$$

Rmk: Log-Sobolev $\implies$ Poincaré

Proof. Claim reversibility of L

$$\mathbb{E}_\pi [f(-Lg)] = \mathbb{E}_\pi [\nabla f \cdot \nabla g]$$

Indeed.

$$\begin{aligned}
\mathbb{E}_\pi [f \cdot (-Lg)] &= \mathbb{E}_\pi [\nabla f \cdot \nabla g] = \mathbb{E}_\pi [f \cdot (-(\Delta g - \nabla V \cdot \nabla g))] \\
&= \int_{\mathbb{R}^d} \left(\frac{1}{Z} e^{-V} f \right) (\nabla V \cdot \nabla g - \Delta g) dx \\
&= \int_{\mathbb{R}^d} \nabla \left(\frac{1}{Z} e^{-V} f \right) \cdot \nabla g dx + \frac{1}{Z} e^{-V} \nabla f \cdot \nabla g dx \\
&= \mathbb{E}_\pi [\nabla f \cdot \nabla g]
\end{aligned}$$

Then by claim, we have

$$\begin{aligned}
\frac{d}{dt} H(X_t \parallel \pi) &= \frac{d}{dt} \int \log \left(\frac{p_t}{\pi} \right) dp_t \\
&= \frac{d}{dt} \mathbb{E}_\pi [(\log h_t) h_t] \\
&= 2 \mathbb{E}_\pi [(\log h_t) L h_t] \\
&= -2 \mathbb{E}_\pi [\nabla(\log h_t) \cdot \nabla h_t] = -2 \mathbb{E}_\pi [|\nabla h_t|^2 / h_t] \\
&= -8 \mathcal{E}(\sqrt{h_t}, \sqrt{h_t}) \leq -16\lambda H(X_t \parallel \pi)
\end{aligned}$$

□

9.4 Inequalities about entropy

In this section, we give 3 inequalities: **Entropy inequality (Donsker-Varadhan)**; **Data processing inequality**; **Pinsker's inequality**.

Theorem 9.4.1 (Entropy inequality (Donsker-Varadhan)).

$$H(\mathbb{P} \parallel \mathbb{Q}) = \sup_f \left\{ \int_{\mathcal{X}} f d\mathbb{P} - \log \left(\int_{\mathcal{X}} e^f d\mathbb{Q} \right) \right\}$$

Proof. Define $\frac{d\mathbb{R}}{d\mathbb{Q}} = \frac{e^f}{Z}$, where normalization factor $Z = \int e^f d\mathbb{Q}$.

" $\geq$ ": By $0 \leq H(\mathbb{P} \parallel \mathbb{R})$, we know that

$$\begin{aligned}
0 \leq H(\mathbb{P} \parallel \mathbb{R}) &= \int \log \left(\frac{d\mathbb{P}}{d\mathbb{R}} \right) d\mathbb{P} = H(\mathbb{P} \parallel \mathbb{Q}) - \int \log \left(\frac{d\mathbb{R}}{d\mathbb{Q}} \right) d\mathbb{P} \\
\implies H(\mathbb{P} \parallel \mathbb{Q}) &\geq \int \log \left(\frac{e^f}{Z} \right) d\mathbb{P} = \int f d\mathbb{P} - \log \left(\int e^f d\mathbb{Q} \right)
\end{aligned}$$

" $\leq$ ": Take $f = \log \left(\frac{d\mathbb{P}}{d\mathbb{Q}} \right)$, and it's easy to check directly that equality holds.

□

Theorem 9.4.2 (Data processing inequality).

For $\mathbb{P}, \mathbb{Q} \in \mathcal{M}_1(\mathcal{X})$, $T : \mathcal{X} \rightarrow \mathcal{Y}$ measurable mapping under $\mathbb{P}$ and $\mathbb{Q}$, define push-forward measures $\mathbb{P}_T = \mathbb{P} \circ T^{-1}$ and $\mathbb{Q}_T = \mathbb{Q} \circ T^{-1}$. Then

$$H(\mathbb{P}_T \parallel \mathbb{Q}_T) \leq H(\mathbb{P} \parallel \mathbb{Q})$$

Proof. Use the result of Donsker-Varadhan, we get

$$\begin{aligned} \text{LHS} &= \sup_{g \in \mathcal{B}(\mathcal{Y})} \left\{ \int_{\mathcal{Y}} g d\mathbb{P}_T - \log \left(\int_{\mathcal{Y}} e^g d\mathbb{Q}_T \right) \right\} \\ &= \sup_{g \in \mathcal{B}(\mathcal{Y})} \left\{ \int_{\mathcal{X}} (g \circ T) d\mathbb{P} - \log \left(\int_{\mathcal{X}} e^{g \circ T} d\mathbb{Q} \right) \right\} \\ &\leq \sup_{f \in \mathcal{B}(\mathcal{X})} \left\{ \int_{\mathcal{X}} f d\mathbb{P} - \log \left(\int_{\mathcal{X}} e^f d\mathbb{Q} \right) \right\} = \text{RHS} \end{aligned}$$

□

Theorem 9.4.3 (Pinsker's inequality).

$$\|\mathbb{P} - \mathbb{Q}\|_{TV}^2 \leq \frac{1}{2} H(\mathbb{P} \parallel \mathbb{Q})$$

Proof. It suffices to prove that

$$H(\mathbb{P} \parallel \mathbb{Q}) \geq 2 \|\mathbb{P} - \mathbb{Q}\|_{TV}^2$$

□

Lecture 10

Diffusion models

10.1 Forward SDEs

Given $F_t : \mathbb{R}^n \rightarrow \mathbb{R}^n$, $\omega : \mathbb{R}_+ \rightarrow \mathbb{R}_+$, $X, B \in \mathbb{R}^n$, consider

$$dX_t = F(X_t) dt + \sqrt{2W_t^2} dB_t, \quad X_0 \stackrel{d}{\sim} P$$

Theorem 10.1.1 (Anderson 1982).

Fix $T > 0$. There exists another SDE $(Y_t)_{t \in [0, T]}$ such that $(Y_t)_{t \in [0, T]} \stackrel{d}{=} (X_{T-t})_{t \in [0, T]}$.

The application to the diffusion model is as follows:

1. sample $Y_0 \stackrel{d}{\sim} P_T$ (we assume this is easy to sample, with $P_T \approx$ white noise);
2. run $(Y_t)_{t \in [0, T]}$;
3. obtain $Y_T \stackrel{d}{\sim} P$.

Lemma 10.1.2.

The Fokker–Planck equation associated to $(X_t)_{t \in [0, T]}$ is

$$\partial_t P_t = \Delta(W_t^2 P_t) - \nabla(F_t P_t), \quad X_t \stackrel{d}{\sim} P_t$$

Proof. For a test function $\phi \in C_c^\infty(\mathbb{R}^n)$, we have

$$\partial_t \mathbb{E}[\phi(X_t)] = \partial_t \int P_t \phi = \int \partial_t P_t \phi$$

and

$$\begin{aligned} d\phi(X_t) &= \nabla\phi(X_t) dX_t + \frac{1}{2} \sum_{i,j=1}^n \partial_{ij}\phi(X_t) d\langle X^i, X^j \rangle_t \\ &= (\nabla\phi(X_t)F(X_t) + W_t^2\Delta\phi(X_t))dt + \text{terms involving } dB_t \end{aligned}$$

so

$$\begin{aligned} \partial_t \mathbb{E}[\phi(X_t)] &= \int (\nabla\phi \cdot F + W_t^2\Delta\phi) P_t \\ &= \int \phi(-\nabla(FP_t) + W_t^2\Delta P_t) \end{aligned}$$

□

Lemma 10.1.3.

The Fokker–Planck equation can be reformulated as

$$\begin{cases} \partial_t P_t &= \nabla(V_t P_t) \\ V_t &= W_t^2 \nabla \log(P_t) - F_t. \end{cases}$$

Proof. We check the right-hand side.

$$\begin{aligned} \nabla(V_t P_t) &= \nabla((W_t^2 \nabla \log(P_t) - F_t) P_t) \\ &= \nabla((W_t^2 \frac{\nabla P_t}{P_t} - F_t) P_t) \\ &= \nabla(W_t^2 \nabla P_t - F_t P_t) = W_t^2 \Delta P_t - \nabla(F_t P_t) \end{aligned}$$

□

Lemma 10.1.4 (ODE viewpoint).

Let $x(t) \in \mathbb{R}^n$ satisfy

$$\begin{cases} x'(t) &= -V_t(x(t)) \\ x(0) &\stackrel{d}{\sim} P_t \end{cases}$$

Then $x(t) \stackrel{d}{\sim} P_t$.

Proof. For any $\phi \in C_c^\infty(\mathbb{R}^n)$,

$$\begin{aligned}\partial_t \mathbb{E}[\phi(X_t)] &= \int \phi \partial_t P_t \\ &= \mathbb{E}[\partial_t \phi(X_t)] \\ &= \mathbb{E}[\nabla \phi \cdot (-V_t(x(t)))] \\ &= \int -(\nabla \phi \cdot V_t) P_t = \int \phi \nabla(V_t P_t)\end{aligned}$$

□

10.2 Reverse of time

Define

$$P_t^b := P_{T-t}, \quad V_t^b := -V_{T-t} = F_{T-t} - W_{T-t}^2 \nabla \log(P_{T-t}).$$

Lemma 10.2.1.

P_t^b solves

$$\begin{cases} \partial_t P_t^b = \nabla(V_t^b P_t^b), & t \in [0, T] \\ P_0^b = P_T. \end{cases}$$

Proof. $P_0^b = P_T$ is obvious. Moreover,

$$\begin{aligned}\partial_t P_t^b &= \partial_t(P_{T-t}) = -\partial_t P_{T-t} \\ &= -\nabla(V_{T-t} P_{T-t}) \\ &= \nabla(V_t^b P_t^b)\end{aligned}$$

□

Lemma 10.2.2.

If

$$\begin{cases} y'(t) &= -V_t^b(y(t)) \\ y(0) &\stackrel{d}{\sim} P_t^b \end{cases}$$

then $y(t) \stackrel{d}{\sim} P_t^b$.

Proof. For any $\phi \in C_c^\infty(\mathbb{R}^n)$, let q_t be the density of $y(t)$. Then

$$\begin{aligned}\partial_t \mathbb{E}[\phi(y(t))] &= \int \phi \partial_t q_t \\ &= \mathbb{E}[\nabla \phi \cdot y'(t)] = \mathbb{E}[-\nabla \phi \cdot V_t^b(y(t))] \\ &= \int -\nabla \phi \cdot V_t^b q_t\end{aligned}$$

□

Lemma 10.2.3.

The Fokker–Planck equation of P_t^b is

$$\begin{cases} \partial_t P_t^b = W_{T-t}^2 \Delta P_t^b - \nabla(U_t^b P_t^b) \\ U_t^b := 2W_{T-t}^2 \nabla \log(P_t^b) - F_{T-t} \end{cases}$$

with associated SDEs

$$\begin{cases} dY_t = U_t^b(Y_t) dt + \sqrt{2W_{T-t}^2} dB_t \\ Y_0 \sim P_T. \end{cases}$$

Proof. The right-hand side of the Fokker–Planck equation is

$$\begin{aligned}\nabla(W_{T-t}^2 \nabla P_t^b - U_t^b P_t^b) \\ &= \nabla((W_{T-t}^2 \frac{\nabla P_t^b}{P_t^b} - U_t^b) P_t^b) \\ &= \nabla((W_{T-t}^2 \nabla \log(P_t^b) - U_t^b) P_t^b)\end{aligned}$$

Thus

$$\begin{aligned}U_t^b &= W_{T-t}^2 \nabla \log(P_t^b) - V_t^b \\ &= W_{T-t}^2 \nabla \log(P_t^b) - (F_{T-t} - W_{T-t}^2 \nabla \log(P_{T-t})) \\ &= 2W_{T-t}^2 \nabla \log(P_t^b) - F_{T-t}.\end{aligned}$$

□

Example 10.2.4 (Ornstein–Uhlenbeck process).

Take $X_t, B_t \in \mathbb{R}^n$ with $dX_t = -X_t dt + \sqrt{2} dB_t$. Observe that if there is no B_t term, $\dot{X}_t =$

$-X_t \Rightarrow (e^t X_t)' = 0 \Rightarrow e^t X_t = X_0$. Consider $Y_t = e^t X_t$. Apply Itô's formula to Y_t :

$$\begin{aligned} dY_t &= e^t X_t dt + e^t dX_t \\ &= e^t X_t dt + (-e^t X_t)dt + \sqrt{2}e^t dB_t \\ &= \sqrt{2}e^t dB_t \end{aligned}$$

Hence $Y_t = Y_0 + \int_0^t \sqrt{2}e^s dB_s$, and $X_t = e^{-t}Y_t = e^{-t}X_0 + e^{-t} \int_0^t \sqrt{2}e^s dB_s$. We have

$$e^{-t} \int_0^t \sqrt{2}e^s dB_s \sim \mathcal{N}(0, \sigma^2)$$

where

$$\begin{aligned} \sigma^2 &= \mathbb{E} \left[\left(e^{-t} \int_0^t \sqrt{2}e^s dB_s \right)^2 \right] \\ &= 2e^{-2t} \int_0^t e^{2s} ds \\ &= 2e^{-2t} \frac{1}{2} (e^{2t} - 1) = 1 - e^{-2t} \end{aligned}$$

Therefore

$$X_t \stackrel{d}{\sim} e^{-t} X_0 + \mathcal{N}(0, I_d(1 - e^{-2t})).$$

10.3 Score matching trick

$\nabla \log(P_t)$ is called the *score*.

Definition 10.3.1.

$$\text{Fisher}(\rho_1 \parallel \rho_2) = \int \rho_1 |\nabla \log(\rho_1) - \nabla \log(\rho_2)|^2 = \mathbb{E}_{X \sim \rho_1} \left[|\nabla \log \rho_1(X) - \nabla \log \rho_2(X)|^2 \right]$$

Lemma 10.3.2.

If s is smooth, then

$$\mathbb{E}_{X \sim P} [|\nabla \log P(X) - s(X)|^2] = \mathbb{E}_{X \sim P} [|s(X)|^2 + 2\nabla \cdot s(X)] + c$$

Proof.

$$\text{LHS} = c + \mathbb{E}_{X \sim P} [|s(X)|^2 - 2s(X) \cdot \nabla \log P(X)]$$

It suffices to treat

$$-2 \mathbb{E}_{X \sim P} [s(X) \cdot \nabla \log P(X)] = -2 \int p s \cdot \nabla \log p = -2 \int p s \cdot \frac{\nabla p}{p} = 2 \mathbb{E}_{X \sim P} [\nabla \cdot s(X)]$$

□

$$\begin{aligned}
S_\theta &= \operatorname{argmin}_{\theta_0 \in \mathbb{R}_0} \mathbb{E} \left[|S_\theta(x)|^2 + 2\nabla \cdot S_\theta(x) \right] \\
&\approx \operatorname{argmin} \frac{1}{n} \sum_{i=1}^n \left[|S_\theta(x_i)|^2 + 2\nabla \cdot S_\theta(x_i) \right]
\end{aligned}$$

Lemma 10.3.3 (Denoising score matching).

Let $X \sim P$, $\epsilon \sim q$, $Y = X + \epsilon$. Then

$$\mathbb{E} \left[|\nabla \log P(Y) - S(Y)|^2 \right] = c + \mathbb{E} \left[|\nabla \log g(\epsilon) - S(Y)|^2 \right]$$

(Can be designed.)

$$\text{DSM: } S_\theta \approx \operatorname{argmin} \frac{1}{n} \sum_{i=1}^n \left[|\nabla \log g(\epsilon) - S(Y)|^2 \right]$$

Proof.

$$\begin{aligned}
-2 \mathbb{E} \left[\nabla \log P_\epsilon(y) \cdot S(y) \right] &= -2 \int P_\epsilon (\nabla \log P_\epsilon \cdot S) dy \\
&= -2 \int \nabla P_\epsilon \cdot S \\
&= -2 \int \nabla \left(\int P(y) q(x-y) dy \right) \cdot S(x) dx \\
&= -2 \iint P(y) \nabla q(x-y) \cdot S(x) dx dy \\
&= -2 \iint P(y) g(x-y) \nabla \log(g(x-y)) \cdot S(x) dx dy \\
&= -2 \iint P(y) g(z) \nabla \log(g(z)) \cdot S(y+z) dz dy \\
&= -2 \mathbb{E} \left[\nabla \log(g(\epsilon)) \cdot S(Y) \right]
\end{aligned}$$

□

Theorem 10.3.4 (Tweedie's formula).

Let $\epsilon \sim \mathcal{N}(0, \sigma^2 I_d)$. Then

$$\nabla \log(P_\epsilon(Y)) = -\mathbb{E} \left[\frac{\epsilon}{\sigma^2} \middle| Y \right] \quad \text{noise prediction formula}$$

$$\nabla \log(P_\epsilon(Y)) = \frac{\mathbb{E}[X | Y] - Y}{\sigma^2} \quad \text{data prediction formula}$$

In summary, what we do in diffusion model is the following;

1. Add noise (e.g. Gaussian) to original data X_0 and get X_t .
2. X_t as training data gives the score $\nabla \log P_t$.
3. When doing actual sampling, do reverse process (solve reverse SDE) on Y_t from a noisy data and sample the original data $p_0 \sim Y_T$.

Lecture 11

JKO Schema

11.1 Flow (Gradient Flow)

Consider the gradient flow:

$$\dot{X}(t) = -\nabla V(X(t))$$

where mappings $t \mapsto X(t) : \mathbb{R}_+ \rightarrow \mathbb{R}^d$ and $x \mapsto V(x) : \mathbb{R}^d \rightarrow \mathbb{R}$.

Key Questions:

Q1: Numerical implementation?

Q2: Generalization in abstract space?

11.1.1 Numerical Implementation

Let $t_k = kh$, where h is the step size, then we have two algorithms:

$$\begin{cases} \text{Forward algorithm (Explicit Euler):} & \frac{X(t_{k+1}) - X(t_k)}{h} = -\nabla V(X(t_k)) \\ \text{Backward algorithm (Implicit Euler):} & \frac{X(t_{k+1}) - X(t_k)}{h} = -\nabla V(X(t_{k+1})) \end{cases}$$

The forward form is what we often use in standard gradient descent.

Lemma 11.1.1.

$X(t_{k+1})$ via the backward algorithm has a variational characterization:

$$X(t_{k+1}) = \operatorname{argmin}_{x \in \mathbb{R}^d} \left\{ \frac{1}{2h} \|x - X(t_k)\|^2 + V(x) \right\} \quad (\text{if the arg exists and is unique})$$

Proof. We want to show that $\frac{x - X(t_k)}{h} = -\nabla V(x)$ if and only if this characterization holds for the backward algorithm. Let $J(y) = \frac{1}{2h} \|y - X(t_k)\|^2 + V(y)$.

- ① If $x = \operatorname{argmin} J(y)$, then for any ε and perturbation vector y , $J(x) \leq J(x + \varepsilon y)$. Since

$$J(x + \varepsilon y) = \frac{1}{2h} \|x - X(t_k) + \varepsilon y\|^2 + V(x + \varepsilon y) = J(x) + \frac{\varepsilon^2}{2h} \|y\|^2 + \frac{\varepsilon}{h} (x - X(t_k)) \cdot y + \varepsilon \nabla V(x) \cdot y + \mathcal{O}(\varepsilon^2)$$

Let $\varepsilon \rightarrow 0$, looking at the first-order variation terms, we get:

$$\frac{1}{h} (x - X(t_k)) \cdot y + \nabla V(x) \cdot y = 0 \implies \frac{x - X(t_k)}{h} = -\nabla V(x)$$

- ② If on the other hand the equation holds, by the same calculation we also get x is a **stationary point** (a point where the derivative is 0) of J .

□

11.2 Heat Equation

Consider the heat equation, i.e. let P be a function on $\mathbb{R}_+ \times \mathbb{R}^d$ satisfying

$$\partial_t P_t = \Delta P_t$$

We also have two numerical algorithms by doing discretization of differential operators:

- **Forward algorithm:**

$$\frac{P(t_{k+1}, x) - P(t_k, x)}{\Delta t} = \frac{1}{\Delta x^2} \sum_{i=1}^d \{P(t_k, x + \Delta x e_i) + P(t_k, x - \Delta x e_i) - 2P(t_k, x)\} \triangleq \Delta_h P(t_k)$$

- **Backward algorithm:**

$$\frac{P(t_{k+1}, x) - P(t_k, x)}{\Delta t} = \frac{1}{\Delta x^2} \sum_{i=1}^d \{P(t_{k+1}, x + \Delta x e_i) + P(t_{k+1}, x - \Delta x e_i) - 2P(t_{k+1}, x)\}$$

Proposition 11.2.1 (Variational description of the algorithm).

The implicit (backward) step can be written as:

$$P(t_{k+1}) = \operatorname{argmin}_P \left\{ \frac{1}{2\Delta t} \|P - P(t_k)\|_{L^2}^2 + \frac{1}{2} \|\nabla_h P\|_{L^2}^2 \right\}$$

where Δ_h is defined above, and $\nabla_h P(t_k) = \frac{P(t_{k+1}) - P(t_k)}{\Delta t}$.

Proof. Again, it suffices to prove that the characterization holds if and only if the equation in the backward algorithm holds.

- ① Suppose P is the argmin of the functional $J(P)$, then for any ε and any perturbation γ , $J(P) \leq J(P + \varepsilon\gamma)$. Since

$$\begin{aligned} J(P + \varepsilon\gamma) &= J(P) + \frac{\varepsilon^2}{2\Delta t} \|\gamma\|_{L^2}^2 + \frac{\varepsilon}{\Delta t} \langle P - P(t_k), \gamma \rangle \\ &\quad + \frac{\varepsilon^2}{2} \|\nabla_h \gamma\|_{L^2}^2 + \varepsilon \langle \nabla_h P, \nabla_h \gamma \rangle \\ &= J(P) + \frac{\varepsilon^2}{2\Delta t} \|\gamma\|_{L^2}^2 + \frac{\varepsilon}{\Delta t} \langle P - P(t_k), \gamma \rangle + \frac{\varepsilon^2}{2} \|\nabla_h \gamma\|_{L^2}^2 - \varepsilon \langle \Delta_h P, \gamma \rangle \end{aligned}$$

Take $\varepsilon \rightarrow 0$:

$$\frac{P - P(t_k)}{\Delta t} - \Delta_h P = 0$$

- ② By the same calculation, P satisfies the backward algorithm equation $\implies P = \operatorname{argmin} J(P)$. ■

□

Lemma 11.2.2.

The problem of the variational formula of the backward algorithm is well-defined.

Proof. Well-definedness here means that there exists a unique solution. Here we shall give a brief idea for the proof. **Existence of solution:** compactness + semicontinuity.

Uniqueness: convexity. □

Proposition 11.2.3.

*The backward algorithm is unconditionally stable, but the forward algorithm requires the **CFL condition**:*

$$\Delta t \leq \frac{1}{2d} |\Delta x|^2$$

for stability to hold.

Proof. Write the two algorithms in vector form:

For forward: $P(t_{k+1}) = (\mathbf{I} + \Delta t \Delta_h) P(t_k)$. Here the matrix is tridiagonal, with diagonal entries $1 - 2|\Delta t|/|\Delta x|^2$, so when $|\Delta t|$ is much larger than $|\Delta x|^2$, the matrix has negative diagonals and hence the algorithm is unstable.

For backward: $(\mathbf{I} - \Delta t \Delta_h) P(t_{k+1}) = P(t_k)$. Since $-\Delta_h$ is positive semi-definite (i.e., $\langle -\Delta_h f, f \rangle = \langle \nabla_h f, \nabla_h f \rangle \geq 0$), the eigenvalues of this matrix satisfy $\lambda_h \leq 0$. Thus $1 - \Delta t \lambda_h \geq 1 \implies |(1 - \Delta t \lambda_h)^{-1}| \leq 1$, proving unconditional stability. □

11.3 JKO Schema (Jordan–Kinderlehrer–Otto) — Another variational formula

Recall that we have introduced the first backward algorithm for the heat equation:

$$P(t_{k+1}) = \operatorname{argmin}_P \left\{ \frac{1}{2\Delta t} \|P - P(t_k)\|_{L^2}^2 + \frac{1}{2} \|\nabla_h P\|_{L^2}^2 \right\} \quad (\text{“Energy”})$$

To rid the algorithm of “differentiation” operations, we hereby introduce a second backward algorithm: the **JKO backward algorithm**. Instead of the L^2 metric, JKO introduces a gradient flow with respect to the **Wasserstein metric** on the space of probability measures $\mathcal{P}_2(\mathbb{R}^d)$:

Definition 11.3.1 (JKO algorithm (for heat equation)).

$$P(t_{k+1}) = \operatorname{argmin}_{P \in \mathcal{P}_2(\mathbb{R}^d)} \left\{ \frac{1}{2h} \mathcal{W}_2^2(P, P(t_k)) + \mathcal{H}(P) \right\}$$

where:

- $\mathcal{H}(P) = \text{Boltzmann entropy} = \int \rho \log \rho \, dx$.
- $\mathcal{W}_2 = 2\text{-Wasserstein distance}$.
- $\mathcal{W}_2^2(\mu, \nu) = \inf_{X \sim \mu, Y \sim \nu} \mathbb{E}[\|X - Y\|^2]$.

Now we prove this is indeed an algorithm for the heat equation.

Proof. Assume P is a minimizer of J (the functional in JKO). Take $\int q = 0$, then:

$$\begin{aligned} J(P) &\leq J(P + \varepsilon q) = \frac{1}{2h} \mathcal{W}_2^2(P + \varepsilon q, P(t_k)) + \mathcal{H}(P + \varepsilon q) \\ &= \frac{1}{2h} \mathcal{W}_2^2(P + \varepsilon q, P(t_k)) + \int (P + \varepsilon q) [\log(P + \varepsilon q)] \end{aligned}$$

Let T be the optimal transport map $T_\# P = P(t_k)$, then:

$$\mathcal{W}_2^2(P + \varepsilon q, P(t_k)) \approx \int |T(x) - x|^2 d(P + \varepsilon q) = \mathcal{W}_2^2(P, P(t_k)) + \varepsilon \int |T(x) - x|^2 dq$$

So

$$J(P) \leq J(P) + \varepsilon \int |T(x) - x|^2 dq + \varepsilon \int q \log P + \varepsilon \int q + \mathcal{O}(\varepsilon^2)$$

By $\int q = 0$, take $\varepsilon \rightarrow 0$, we get the optimality condition:

$$\int |T(x) - x|^2 q + \int q \log P = 0$$

which leads to the heat equation by more advanced methods. $\square$

Finally, we give a more general result for Langevin dynamics:

Theorem 11.3.2 (JKO).

Consider the stochastic differential equation:

$$dX_t = -\nabla V(X_t) dt + \sqrt{2\beta^{-1}} dW_t$$

Define the functionals:

$$\mathcal{H}(P) = \int P \log P, \quad \mathcal{E}(P) = \int V dP, \quad \mathcal{F}(P) = \mathcal{E}(P) + \mathcal{H}(P) \cdot \beta^{-1}$$

Then the discrete JKO step:

$$P^{(h)}(t_{k+1}) = \operatorname{argmin}_{P \in \mathcal{P}_2(\mathcal{M})} \left\{ \frac{1}{2h} \mathcal{W}_2^2(P, P^{(h)}(t_k)) + \mathcal{F}(P) \right\}$$

When $h \rightarrow 0$, $P^{(h)}(t) \rightarrow P(t)$ weakly in L^1 , where $P(t)$ is the distribution for the solution X_t of the equation.

Remark 11.3.3.

We can use the JKO scheme to define the heat equation or a Langevin dynamics on a manifold / metric space. A natural question is: Does this construction coincide with the other constructions?